CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – I

CEMENTS & AGGREGATES

Syllabus covered in this unit

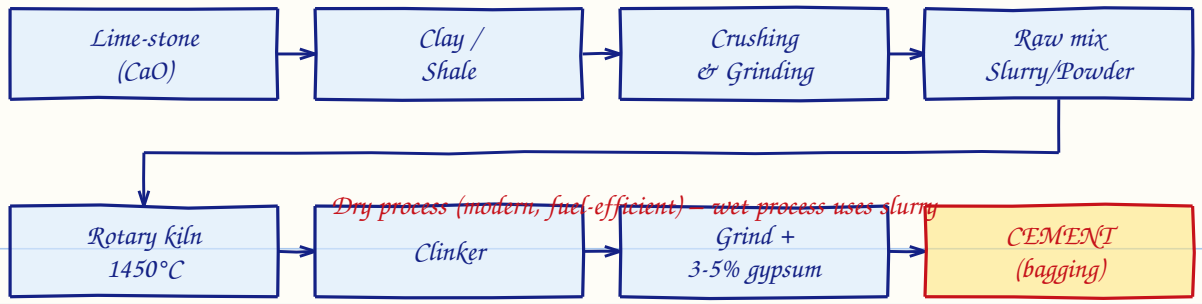
Portland cement – chemical composition – hydration – setting – fineness – structure of hydrated cement – physical tests – grades of cement – admixtures (mineral & chemical) – classification of aggregates – shape, texture, bond, strength – specific gravity, bulk density, porosity, absorption, moisture – bulking – deleterious substances – soundness – alkali-aggregate reaction – thermal properties – sieve analysis – FM – grading – max size – mixing water

1. PORTLAND CEMENT

Cement = a **finely ground hydraulic binder**; when mixed with water it sets and hardens by chemical reaction (hydration) – even under water. Joseph Aspdin (1824) patented it & named it "Portland" cement because it resembled Portland stone.

1.1 Manufacture (brief)

- Raw materials : **calcareous** (limestone, chalk) + **argillaceous** (clay, shale) + gypsum (added at the end).
- Proportion about 3 : 1 (lime : clay) – burnt in a rotary kiln at about 1400–1450 °C → **clinker** (nodules 3–25 mm).
- Clinker is cooled, ground with **3–5 % gypsum** to control the setting of C_3A.



1.2 Chemical composition (oxide composition)

Oxide	Per cent	Function / remark
Lime CaO	60 – 67	Main ingredient; excess free lime → unsoundness; deficiency → low strength, quick set
Silica SiO_2	17 – 25	Gives strength (C_3S, C_2S)

Oxide	Per cent	Function / remark
Alumina Al_2O_3	3 – 8	Quick setting, lowers clinkering temp.
Iron oxide Fe_2O_3	0.5 – 6	Colour, helps fusion
Magnesia MgO	0.1 – 4	Excess (>6 %) → unsoundness
Sulphur trioxide SO_3	1 – 3	Excess → unsoundness (sulphate attack)
Alkalis $Na_2O + K_2O$	0.4 – 1.3	Excess → efflorescence, alkali-aggregate reaction

- IS 269 limits : insoluble residue ≤ 4 %, MgO ≤ 6 %, loss on ignition ≤ 5 %, SO_3 ≤ 2.5 % (C_3A ≤ 5 %) or 3 % (C_3A > 5 %), chloride ≤ 0.1 %.
- **Lime Saturation Factor** $LSF = (CaO - 0.7\ SO_3) / (2.8\ SiO_2 + 1.2\ Al_2O_3 + 0.65\ Fe_2O_3)$ → 0.66 to 1.02. **Alumina–iron ratio** = Al_2O_3/Fe_2O_3 ≥ 0.66.

1.3 Bogue's compounds (main compounds of cement)

Compound	Formula (short)	%	Behaviour
Tricalcium silicate C_3S	$3CaO{\cdot}SiO_2$	25–50	Hydrates fast, high heat (500 J/g); early strength (7 days) & final set
Dicalcium silicate C_2S	$2CaO{\cdot}SiO_2$	20–45	Slow; low heat (260 J/g); later strength after 28 days
Tricalcium aluminate C_3A	$3CaO{\cdot}Al_2O_3$	5–11	Flash set; highest heat (910 J/g); poor sulphate resistance
Tetra-calcium alumino-ferrite C_4AF	$4CaO{\cdot}Al_2O_3{\cdot}Fe_2O_3$	6–12	Colour (grey); low heat (420 J/g); little strength

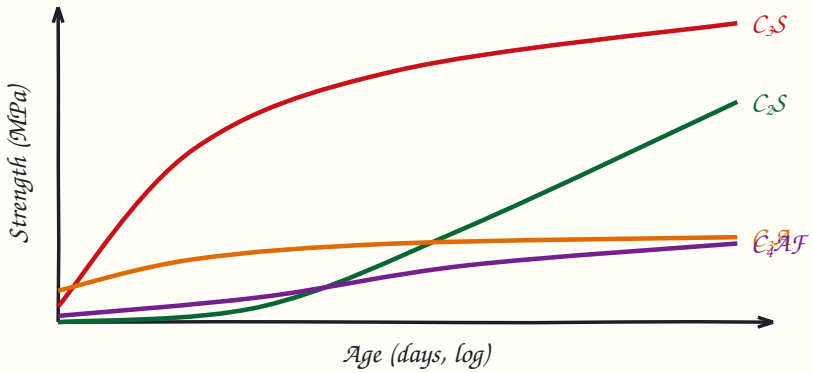
Bogue's equations (when $A/F \geq 0.66$) $C = \%CaO$, $S = \%SiO_2$, $A = \%Al_2O_3$, $F = \%Fe_2O_3$

$C_3S = 4.07\ C - 7.60\ S - 6.72\ A - 1.43\ F - 2.85\ SO_3$

$C_2S = 2.87\ S - 0.754\ C_3S$

$C_3A = 2.65\ A - 1.69\ F$

$C_4AF = 3.04\ F$



- C_3S : early strength
- C_2S : later strength
- C_3A : flash set, heat, little strength
- C_4AF : colour, little strength

2. HYDRATION OF CEMENT

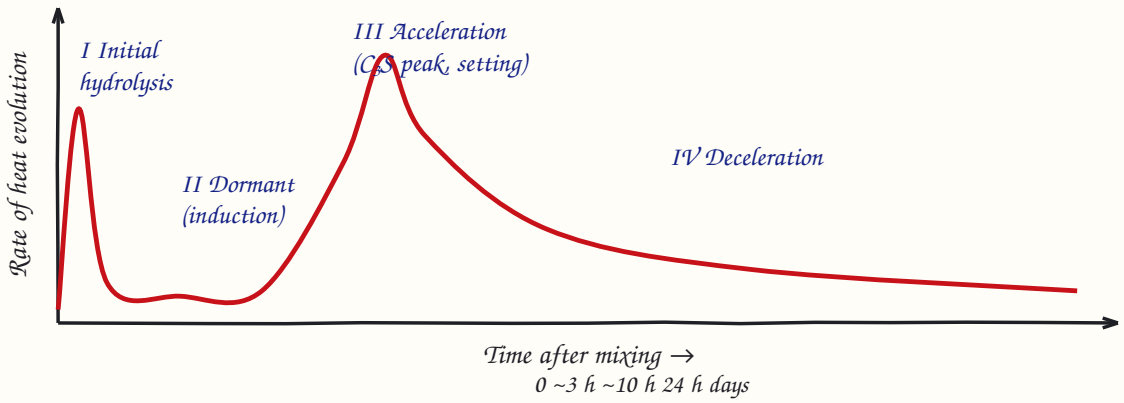
Hydration = chemical reaction between cement compounds and water forming **hydration products (gel)** that bind the aggregates. It is **exothermic**.

- $2C_3S + 6H \rightarrow C_3S_2H_3$ (C-S-H gel) + $3Ca(OH)_2$ [100 : 24 → 75 : 49 by weight]
- $2C_2S + 4H \rightarrow C_3S_2H_3 + Ca(OH)_2$ (less lime, hence more durable)
- $C_3A + 6H \rightarrow C_3AH_6$ (very fast, flash set if no gypsum)
- $C_3A + 3CaSO_4 \cdot 2H_2O + 26H \rightarrow$ **ettringite** ($C_3A \cdot 3CaSO_4 \cdot 32H_2O$) → later converts to monosulphate.
- $C_4AF + CaSO_4 + H \rightarrow$ calcium sulpho-aluminoferrite + CH + hydrates.

Remember ★

- Water needed for complete hydration ≈ **23 %** (chemically bound) + **15 %** (gel water) ≈ **38 %** by weight of cement → minimum w/c ≈ 0.38.
- $Ca(OH)_2$ (CH) gives alkalinity (pH ≈ 12.5) which protects steel but is weak & soluble (leaching).
- Heat of hydration : 7 days ≈ 250–300 J/g, 28 days ≈ 310–350 J/g → problem in mass concrete.

2.1 Stages of hydration (heat evolution curve)



- Stage I – rapid dissolution, C_3A & gypsum react (ettringite film) – lasts few minutes.
- Stage II – **dormant / induction period** (1–3 h) : paste stays plastic → time for transport & placing.
- Stage III – acceleration : C_3S & C_2S hydrate fast, **initial & final set** occur, CH crystallises.
- Stage IV/V – deceleration, diffusion controlled, slow continued strength gain for months.

2.2 Setting of cement

- **Setting** = change from fluid to rigid state (no strength yet); **hardening** = gain of strength after setting.

- **Initial set** : paste starts losing plasticity (≥ 30 min as per IS) ; **final set** : paste fully rigid, can bear some load (≤ 600 min).
- **Flash set** : C_3A reacts instantly with water – stiffening with lot of heat, can't be reworked; prevented by gypsum.
- **False set** : abnormal premature stiffening without much heat (dehydration of gypsum to hemi-hydrate); regained on remixing without extra water.
- Factors : cement composition (C_3A), fineness, w/c, temperature (high → faster), gypsum content, admixtures.

3. FINENESS OF CEMENT

- Fineness controls **rate of hydration** : finer cement → more surface → faster set, early strength, more heat, more shrinkage; but coarse particles (>45 μm) may never fully hydrate.
- Typical OPC : specific surface 300–400 m^2/kg (IS 269 minimum 225 m^2/kg).

Methods of determining fineness

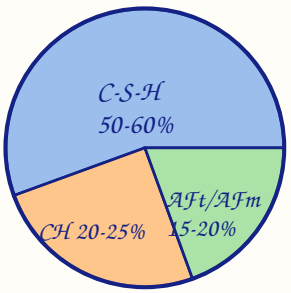
1. **Sieve test** (IS 4031 Part 1) – 100 g cement on 90 μm IS sieve, sieved 15 min; residue should be ≤ 10 %.
2. **Air permeability (Blaine's) method** – specific surface from time for fixed air volume to flow through a cement bed of known porosity : $S \propto \sqrt{(e^3)} \cdot \sqrt{t} / \rho(1-e)$.
3. **Wagner turbidimeter** – sedimentation of particles in kerosene, light transmission gives particle-size distribution.
4. **Lea & Nurse permeability apparatus**, laser diffraction (modern).

Point ★

Finer cement → greater **bleeding control**, better workability, but more **heat & shrinkage**, and faster air-setting (deterioration in storage).

4. STRUCTURE OF HYDRATED CEMENT PASTE

- Hydrated paste = **C-S-H gel** (tobermorite-like, poorly crystalline, 50–60 % of solids, specific surface 200–300 m^2/g) + **$Ca(OH)_2$** crystals (20–25 %) + **sulpho-aluminates** (AFt, AFm 15–20 %) + unhydrated clinker + pores.
- Strength arises from **Van der Waals' forces** (physical bonding) between gel particles; gel has very large surface.



Solids of hydrated paste

Pores in hydrated paste :

- Gel pores ≈ 0.5 – 2.5 nm (inside C-S-H)
- Capillary pores ≈ 10 nm – 50 µm (space left by excess mix water)
- Air voids : entrapped (>1 mm) / entrained (50 µm – 1 mm)

Low w/c → fewer capillary pores → higher strength

Types of water in the paste (Powers–Brunauer model)

- **Chemically combined water** (non-evaporable) – part of hydrates; lost only at high temp.
- **Gel (interlayer) water** – adsorbed on gel surface, lost on drying below ~11 % RH; causes **creep & shrinkage**.
- **Capillary water** – free water in capillary pores (>50 nm): removal causes little shrinkage.
- **Free / evaporable water** – in large voids; evaporates easily.

Powers' relations (gel/space ratio) – used again in Unit III

Gel/space ratio $X = 0.657\,\alpha / (0.319\,\alpha + w/c)$ Strength $f = 240\,X^3$ (MPa)

α = degree of hydration; gel volume = 2.06 × volume of cement hydrated

5. TESTS FOR PHYSICAL PROPERTIES (IS 4031)

Test	Procedure / result
1. Fineness	Sieve test (90 µm, ≤ 10 %) / Blaine (≥ 225 m^2/kg)
2. Standard consistency	**Vicat apparatus**, plunger 10 mm dia, 300 g. Water % (P) at which plunger stops **5–7 mm from bottom** (33–35 mm penetration). Normal P ≈ 26–33 %.
3. Initial setting time	Vicat with **1 mm² square needle**; water = 0.85 P; time when needle stops **5 ± 0.5 mm from base**. ≥ 30 min.
4. Final setting time	Needle with **annular collar (5 mm)** : needle makes an impression but collar doesn't. ≤ 600 min.
5. Soundness	**Le Chatelier** (mould 30 mm × 30 mm, split cylinder, arms 165 mm) : water 0.78 P, 24 h curing, boil 3 h → expansion ≤ **10 mm**. **Autoclave** test ≤ 0.8 %.
6. Compressive strength	Mortar 1 : 3 (cement : standard sand, IS 650), water = (P/4 + 3.0) %, **70.6 mm cubes**; tested at 3, 7, 28 days.
7. Specific gravity	Le Chatelier flask, kerosene ; **3.10 – 3.15**.
8. Heat of hydration	Calorimeter ; for low heat cement 7 d ≤ 272 kJ/kg.

Test	Procedure / result
9. Tensile / Soundness extras	Briquette test (obsolete) ; Setting & expansion due to MgO, free lime, SO_3.

Standard consistency – Vicat ★

Plunger : 10 mm dia × 50 mm long, mass 300 g. Mould : 40 mm deep, 80–90 mm dia. Trials of 300 g cement + 25 %, 28 %, 30 % … water till plunger rests **5–7 mm above the base**.

6. GRADES AND TYPES OF CEMENT

6.1 Grades of OPC (IS 269 / 8112 / 12269)

Grade (28-day strength)	3-day (MPa)	7-day (MPa)	28-day (MPa)
33 grade (IS 269)	≥ 16	≥ 22	≥ 33
43 grade (IS 8112)	≥ 23	≥ 33	≥ 43
53 grade (IS 12269)	≥ 27	≥ 37	≥ 53

- 33 – plastering, masonry ; 43 – general RCC ; 53 – high-rise, prestressed, precast, roads, high-strength concrete.
- Grade number = minimum 28-day compressive strength of 70.6 mm mortar cubes in N/mm^2.

6.2 Other cements

Type (IS code)	Salient features / use
Rapid Hardening (IS 8041)	More C_3S, finer; 3-day strength = OPC 7-day; early de-shuttering, repairs, cold weather
Low Heat (IS 12600)	Less C_3A (<6 %) & C_3S, more C_2S; dams, mass concrete
Sulphate Resisting (IS 12330)	C_3A ≤ 5 %; marine works, sewers, foundations in sulphate soils
Portland Pozzolana PPC (IS 1489)	15–35 % fly ash (Part 1) / calcined clay (Part 2); low heat, better durability, slow early strength
Portland Slag PSC (IS 455)	25–70 % GGBS; low heat, sulphate & chloride resistance, marine works
White / Coloured (IS 8042)	Low iron; decorative, floor finishes, architectural
High Alumina (IS 6452)	35–40 % alumina; very early strength, refractory; not for structural use due to conversion
Hydrophobic (IS 8043)	Water-repellent film on grains; long storage, monsoon
Oil-well, Expansive, Masonry (IS 3466)	special uses

7. ADMIXTURES

Admixture = material other than cement, water, aggregates added **immediately before or during mixing** to modify properties of fresh / hardened concrete (IS 9103, ASTM C494). Used ≤ 5 % by wt. of cement.

7.1 Chemical admixtures

Type	Examples	Effect / dosage
Accelerators	$CaCl_2$ (≤ 1.5–2 %), triethanolamine, sodium silicate, calcium formate	Fast set & early strength; cold weather. **$CaCl_2$ not for RCC** (corrosion), increases shrinkage
Retarders	Sugar, gypsum, calcium sulphate, starch, lignosulphonates, hydroxy-carboxylic acids	Delay setting, reduce heat; hot weather, long haul RMC, large pours
Air-entraining	Vinsol resin, vegetable oils/fats, fatty acids, abietic acid salts, detergents	3–6 % tiny (50 µm) bubbles; better freeze–thaw, workability, less bleeding; **strength drops ≈ 5 % per 1 % air**
Plasticizers (water reducers)	Lignosulphonates, hydroxylated carboxylic acids	5–15 % water reduction; dose 0.1–0.4 %
Super-plasticizers (HRWR)	SNF (naphthalene), SMF (melamine), modified lignosulphonate, **PCE (polycarboxylate)**	20–30 % (PCE up to 40 %) water cut or flowing concrete (slump 150→ 250 mm); dose 0.5–2 %; slump loss in 30–60 min

- Mechanism of plasticizers : adsorb on cement grains → **deflocculation** by electrostatic (SNF/SMF) or **steric** (PCE) repulsion → trapped water released.
- Other : VMA, shrinkage-reducing, corrosion inhibitors, waterproofing, bonding, gas-forming (aluminium powder), pumping aids.

7.2 Mineral admixtures (supplementary cementitious materials)

- **Pozzolana** = siliceous/alumino-siliceous material which by itself has no cementing value but, finely divided and with moisture, reacts with $Ca(OH)_2$ at ordinary temperature → C-S-H : $CH + S + H \rightarrow$ **C-S-H** (secondary gel).

Material	Properties	Effect on concrete
Fly ash (IS 3812)	Residue of pulverised coal, spherical glassy 1–100 µm; Class F (siliceous, $SiO_2+Al_2O_3+Fe_2O_3 \geq 70$ %), Class C (calcareous, self-cementing). Replace 15–35 %	Better workability (ball-bearing effect), low heat, durability, sulphate & ASR resistance; slower early strength, needs longer curing

Material	Properties	Effect on concrete
Silica fume (IS 15388)	By-product of Si / Fe-Si alloy; SiO_2 > 85–90 %; particle size ≈ 0.1 µm (100× finer than cement); 15 000–25 000 m^2/kg; 5–10 %	Filler + highly reactive pozzolan → very dense, 80–100+ MPa HPC; reduces bleeding; high water demand → needs superplasticizer
GGBS	Granulated blast-furnace slag, 25–70 %	Low heat, chloride & sulphate resistance
Rice husk ash, metakaolin	Reactive silica	HPC, durability

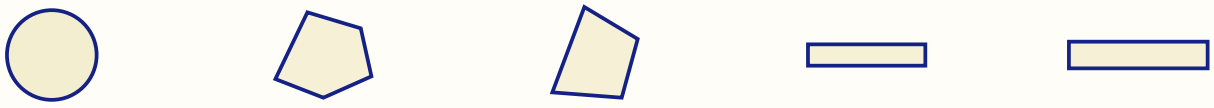
8. AGGREGATES

Aggregates occupy **60–75 % of concrete volume** (70–80 % of mass) – they affect strength, durability, workability, economy, dimensional stability.

8.1 Classification

- **By size** : **Coarse** aggregate – retained on 4.75 mm sieve (nominal 10, 12.5, 20, 25, 40 mm). **Fine** aggregate – passes 4.75 mm, retained on 75 µm (sand). **All-in aggregate** – mix of both.
- **By source / origin** : natural (river, pit, sea, crushed rock – granite, basalt, quartzite, limestone) – artificial (broken brick, slag, clinker, expanded clay).
- **By unit weight** : **Normal weight** (2400 kg/m^3 concrete; bulk density 1520–1680 kg/m^3); **Light-weight** (< 1120 kg/m^3, pumice, vermiculite, expanded shale); **Heavy-weight** (> 2080 kg/m^3; barytes, magnetite – radiation shielding).
- **By shape** : rounded, irregular, angular, flaky, elongated.
- **By geological (petrological) group** : igneous, sedimentary, metamorphic.

9. PARTICLE SHAPE AND TEXTURE



Rounded Irregular Angular Flaky Elongated

- **Rounded** (river gravel) : minimum voids, best workability, poor bond, lower flexural strength.
- **Angular / crushed** : more interlock & bond → higher strength (10–15 % more flexural), but needs ~10 kg more water per m^3; harsh mix.
- **Flaky** (thickness < 0.6 × mean size) & **elongated** (length > 1.8 × mean size) → poor workability, weaker, lie flat with water pockets → limit **combined index ≤ 30–35 %**.
- **Surface texture** : smooth (glassy, chert) ↔ rough (honeycombed, granular) → rough texture gives

better **bond** & strength; smooth gives workability.

Shape tests (IS 2386 Part 1) ★

- **Flakiness index** = (weight passing thickness-gauge slot / total weight) × 100.
- **Elongation index** = (weight retained on length gauge / total) × 100.
- **Angularity number** = 67 − % solid volume of aggregate in a compacted cylinder (0 for rounded, up to 11 for very angular); **Angularity index** = ratio of void % .

10. BOND, STRENGTH & MECHANICAL PROPERTIES

- **Bond** between aggregate and paste (interfacial transition zone, ITZ – weakest link, thickness 15–50 µm) depends on texture, shape, mineralogy, cleanliness. Bond is evaluated indirectly – by flexural strength comparison, no standard test.
- **Strength of aggregate** : should exceed that of concrete (rocks 70–300 MPa); weak aggregate (e.g. pumice) limits strength of concrete.

Test (IS 2386 Part 4)	Procedure	Limits (IS 383)
Aggregate Crushing Value (ACV)	12.5–10 mm aggregate in 150 mm cylinder, load 40 t in 10 min; % fines passing 2.36 mm	≤ 45 % (other concrete); ≤ 30 % wearing surfaces
Aggregate Impact Value (AIV)	10–12.5 mm, 14 kg hammer, 15 blows from 380 mm; % passing 2.36 mm	≤ 45 % ; ≤ 30 % for wearing surfaces
Los Angeles Abrasion	Aggregate + 6–12 steel balls, drum 500 rev (30–33 rpm); % loss passing 1.7 mm	≤ 50 % ; ≤ 30 % for wearing surfaces
Aggregate Abrasion (Deval), 10 % fines value	Rate of wear / load for 10 % fines	– (roads)
Toughness / Hardness	Impact & Abrasion tests	Toughness = resistance to impact; Hardness = resistance to wear

- $ACV = (W_2 / W_1) \times 100$, W_1 = weight of sample, W_2 = weight passing 2.36 mm.

11. PHYSICAL PROPERTIES OF AGGREGATES

11.1 Specific gravity, bulk density, porosity, absorption

- **Specific gravity** (G) = weight of dry aggregate / weight of equal volume of water. Normal rocks : **2.6 – 2.9** (granite 2.65, basalt 2.9). Apparent G (solid only) vs **bulk G (SSD)** (includes pores). Test : pycnometer / wire-basket method.
- **Bulk density** = weight / volume incl. voids: loose 1450–1550, compacted 1600–1750 kg/m^3.

Voids % = $(1 - \gamma / (G \cdot \gamma_w)) \times 100 \approx 35–40$ % for coarse.

- **Porosity** = volume of pores in aggregate / total volume : 0 – 20 %; affects strength, absorption, permeability, freeze-thaw.
- **Absorption** = $(W_{SSD} - W_{dry}) / W_{dry} \times 100$; 24 h immersion. Normal : 0.1 – 2 % ; > 2–3 % considered poor.

11.2 Moisture states of aggregate

State	Description
Oven-dry	All moisture removed (110 °C); absorbs water from mix
Air-dry	Surface dry, some internal moisture
SSD (saturated surface-dry)	Pores full, surface dry – **reference state** (no water exchange)
Damp / wet	Pores full + surface (free) moisture → adds to mixing water

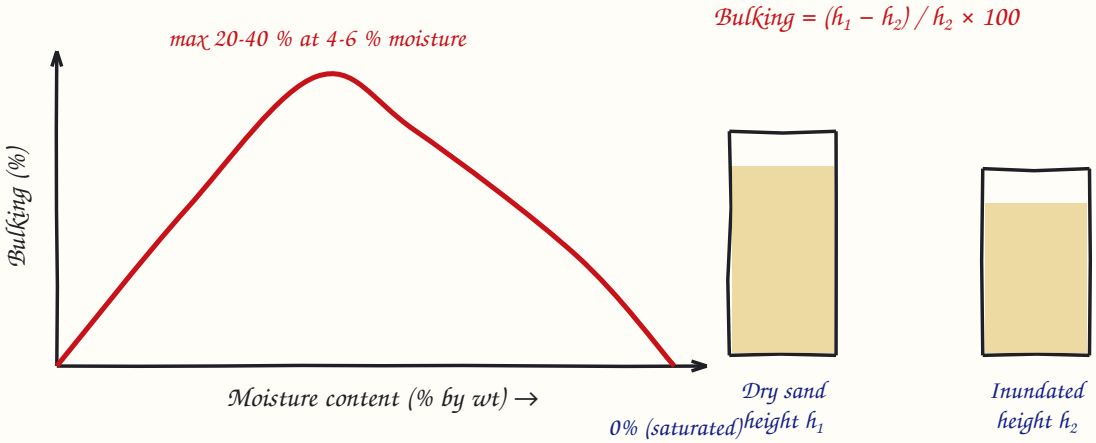
Moisture content & surface water

Moisture content (%) = $(W - W_{dry}) / W_{dry} \times 100$

Free (surface) moisture = Moisture content – Absorption (negative → aggregate absorbs water from the mix)

Correction : water to add = designed water – Σ(free moisture) + Σ(absorbed)

12. BULKING OF SAND



- Thin film of water on sand particles → **surface tension pushes grains apart** → increase in **volume** of moist sand. Fine sand bulks more (up to 40 %) ; coarse sand ≈ 15–20 %.
- Maximum bulking at **4–6 % moisture** ; when completely saturated (submerged) → bulking nil.
- **Field test** : fill a measuring cylinder (250 ml) with moist sand to height h_1 ; add excess water, stir – sand settles to h_2 → bulking % = $(h_1 - h_2)/h_2 \times 100$.
- **Importance** : if volume-batching is used without correcting for bulking, the sand actually

delivered is **less** → harsh, honeycombed, lower-strength mix. Hence quantity of moist sand is increased by the bulking %.

13. DELETERIOUS SUBSTANCES IN AGGREGATE

Impurity	Harmful effect	Limit / test
Organic matter (humus)	Retards hydration, loses strength	Colour test with NaOH — colour must not be darker than standard (straw/yellow)
Clay, silt, fine dust	Increases water demand, shrinkage, coats grains (poor bond)	Max 3 % (natural sand) , 1 % coarse (IS 383); sedimentation / decantation test
Salts (chlorides, sulphates)	Efflorescence, reinforcement corrosion, sulphate attack	Chloride ≤ 0.06 % ; total sulphate ≤ 0.5 % (as SO_3)
Coal, lignite, shale	Swelling, staining, popouts	≤ 1 %
Unsound particles (soft stone, mica, pyrites)	Weakness, rust stains, pop-outs	Max 5 % by wt
Sea-shells, sea salt (sea sand)	Chloride → corrosion	Must be washed

14. SOUNDNESS OF AGGREGATE

- **Soundness** = resistance of aggregate to **disintegration/volume change** caused by weathering – freezing–thawing, wetting–drying, heating–cooling.
- **Test (IS 2386 Part 5)** : sample immersed in saturated Na_2SO_4 (or $MgSO_4$) solution for 16–18 h → oven dried at 105–110 °C → **5 cycles**; salt crystallisation in pores simulates freeze–thaw.
- Limits (IS 383) : weight loss after 5 cycles **≤ 12 %** with sodium sulphate, **≤ 18 %** with magnesium sulphate.

15. ALKALI–AGGREGATE REACTION (AAR)

- Reaction between **alkalis** (Na_2O, K_2O) of cement (and other sources) and certain **reactive minerals** of aggregate in presence of moisture, producing **expansive gel** → cracks, pop-outs, map cracking, loss of strength & durability.
- **Alkali–Silica Reaction (ASR)** – reactive silica (opal, chalcedony, chert, tridymite, cristobalite, volcanic glass) + alkali → **alkali-silicate gel** which absorbs water and swells (pressure > tensile strength).
- **Alkali–Carbonate Reaction** – dolomitic limestone + alkali → de-dolomitisation, brucite.
- **Conditions** : (1) reactive aggregate (2) alkali > 0.6 % (Na_2O eq = Na_2O + 0.658 K_2O) (3) enough

moisture (RH > 80 %).

Control of AAR ★

- Use **non-reactive aggregates** (petrographic test, mortar-bar test ASTM C227, chemical test ASTM C289)
- **Low-alkali cement** (< 0.6 % Na_2O eq) ; use PPC / fly ash / slag / silica fume (they consume alkalis)
- Keep concrete **dry** (prevent moisture ingress), lithium admixtures, limit cement content.

16. THERMAL PROPERTIES OF AGGREGATES

- **Coefficient of thermal expansion** : 5 – 13 × 10^{-6} /°C (quartzite, flint highest ≈ 11–13 ; limestone, basalt ≈ 5–7 ; granite 7–9). Concrete ≈ 10 × 10^{-6} /°C (steel 12 × 10^{-6}) – compatibility of concrete & steel is the reason RCC works.
- Difference between aggregate and paste expansion → internal stress in fire / freeze-thaw cycles. **Specific heat** ≈ 0.18–0.22 cal/g°C ; **thermal conductivity** 1.5 – 3.5 W/m·K ; **diffusivity** governs cooling of mass concrete.
- Low-expansion aggregate (limestone) is preferred for fire resistance & pavements, high-expansion ones cause cracking.

17. SIEVE ANALYSIS, FM & GRADING

- **Grading** = particle size distribution of aggregate, found by **sieve analysis** (IS 2386 Part 1): a weighed sample is passed through sieves of decreasing size arranged in a nest and shaken ; % retained on each sieve is found.
- IS sieves : coarse – **80, 40, 20, 10, 4.75 mm** ; fine – **2.36 mm, 1.18 mm, 600 µ, 300 µ, 150 µ.**
- **Fineness Modulus (FM)** = Σ (cumulative % retained on standard sieves 80 mm … 150 µ) / 100. Higher FM = coarser.

Sieve size	Weight retained (g)	% retained	Cumulative % retained
10 mm	0	0	0
4.75 mm	40	4	4
2.36 mm	100	10	14
1.18 mm	150	15	29
600 µ	250	25	54
300 µ	250	25	79
150 µ	150	15	94

Sieve size	Weight retained (g)	% retained	Cumulative % retained
Pan	60	6	–

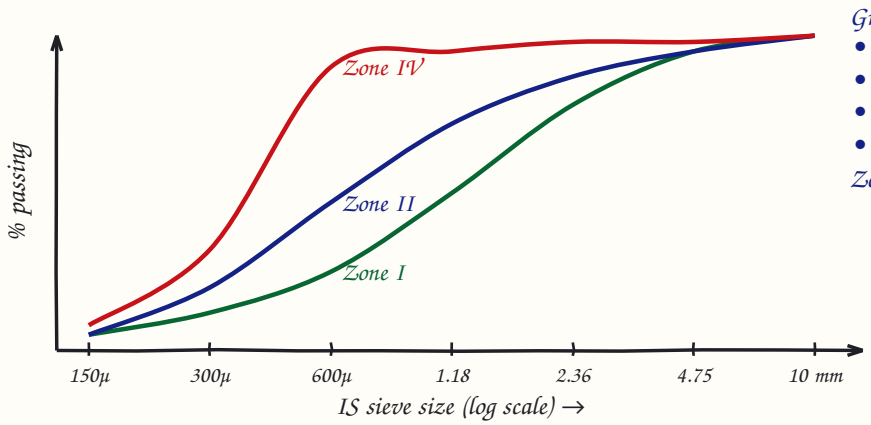
Worked example (1 kg sand) ★

FM = (0 + 4 + 14 + 29 + 54 + 79 + 94) / 100 = **2.74** → medium sand (Zone II).

Fine sand FM 2.2 – 2.6 ; medium 2.6 – 2.9 ; coarse 2.9 – 3.2 ; **coarse aggregate 5.5 – 8.0** ; all-in aggregate 3.5 – 6.5.

Combined FM of blend : $FM = (x{\cdot}FM_1 + y{\cdot}FM_2) / (x + y)$ [or % fine aggregate = $(F_{CA} - F_{blend})/(F_{CA} - F_{FA}) \times 100$]

17.1 Grading curves



Grading curve:
- S-shaped = well graded
- Straight/steep = uniform
- Hump = gap graded
- Zone I coarse → Zone IV fine

▶ **Well graded** (S curve) : all sizes present → min. voids, less paste, best strength & workability.
Gap graded : some intermediate sizes missing (suited for no-fines / pumping with care).
Uniformly graded : single size.

17.2 Grading of fine aggregate (IS 383 : 2016) – % passing

IS sieve	Zone I	Zone II	Zone III	Zone IV
10 mm	100	100	100	100
4.75 mm	90–100	90–100	90–100	95–100
2.36 mm	60–95	75–100	85–100	95–100
1.18 mm	30–70	55–90	75–100	90–100
600 μ	15–34	35–59	60–79	80–100
300 μ	5–20	8–30	12–40	15–50
150 μ	0–10	0–10	0–10	0–15

▶ Zone I (coarsest) → Zone IV (finest); Zone II is the most commonly used; Zones I–III OK for RCC, Zone IV (very fine) needs richer mix/ reduced fine aggregate.

17.3 Grading of coarse aggregate (IS 383) – % passing

IS sieve	40 mm graded	20 mm graded	40 mm single	20 mm single
63 mm	100	–	100	–
40 mm	95–100	100	85–100	100
20 mm	30–70	95–100	0–25	85–100
10 mm	10–35	25–55	0–5	0–20
4.75 mm	0–5	0–10	–	0–5

18. MAXIMUM SIZE OF AGGREGATE (MSA)

- **MSA** = smallest sieve on which ≥ 10 % retained (nominal max size = size of sieve passing 90–100 %).
- Larger MSA → less water & cement (smaller surface area) → economical, less shrinkage; but **limited by** :
  - ≤ ¼ of **minimum thickness / dimension** of member
  - ≤ **5 mm less than minimum clear spacing** between reinforcement bars; ≤ ¾ of cover (IS 456 cl. 5.3.3)
  - 20 mm typical for RCC; 40 mm for mass / PCC; 10–12 mm for thin / heavily reinforced sections.
- Increasing MSA from 20 → 40 mm : strength ↑ only in lean mixes; in rich mixes strength falls (smaller bond area, more bleeding at ITZ).

19. QUALITY OF MIXING WATER (IS 456 : 2000, cl. 5.4)

- Water used for mixing and curing should be **clean, free from oils, acids, alkalis, sugars, salts, organic matter**. **Potable water** is generally satisfactory.
- pH ≥ 6. Acidity : max 2 ml of 0.02 N NaOH to neutralise 100 ml water (phenolphthalein). Alkalinity : max 25 ml of 0.02 N H_2SO_4 (methyl orange).

Impurity (permissible limit, mg/litre)	PCC	RCC
Organic	200	200
Inorganic	3000	3000
Sulphates (as SO_3)	400	400
Chlorides (as Cl)	2000	**500**
Suspended matter	2000	2000

- Comparison test : 28-day strength of cubes with test water ≥ **90 %** of strength with distilled water ; initial setting time not less than 30 min and difference from control ± **30 min**.

- **Sea water** : acceptable for PCC (not for RCC / prestressed) – causes efflorescence, corrosion, dampness, increases early strength but ↓ 28-day strength by ~ 10–15 %.

Quick revision – Unit I ★

- Cement : C_3S (early), C_2S (late), C_3A (flash set, gypsum), C_4AF (colour) ; w/c min ≈ 0.38 ; Le Chatelier ≤ 10 mm ; Initial ≥ 30 min, final ≤ 600 min.
- Admixtures : accelerator ($CaCl_2$), retarder (sugar), AEA (3–6 % air), plasticizer 5–15 %, SP 20–30 %.
- Aggregate : FM sand 2.2–3.2, CA 5.5–8 ; bulking max 4–6 % moisture ; soundness 5 cycles, 12 % / 18 % ; MSA ≤ ¼ min dimension.

CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – II

FRESH CONCRETE

Syllabus covered in this unit

Steps in the manufacture of concrete – proportioning, mixing, placing, compaction, finishing, curing and their types – properties of fresh concrete – workability – factors – tests – setting times – effect of time & temperature – segregation & bleeding – mixing & vibration – ready mixed concrete – shotcrete

1. INTRODUCTION

- **Fresh (green) concrete** = concrete from the time of mixing till it sets (about 2–4 h). Good fresh concrete must be **mixed, transported, placed, compacted and finished easily without segregation**.
- Quality of hardened concrete (strength, durability) depends more on **how concrete is made & handled** than on mix design alone.



Stages in the manufacture of concrete (each stage affects final quality)

Concrete = Cement + Water + Fine aggregate + Coarse aggregate (+ admixtures)★

Typical volume : cement 7–15 %, water 14–21 %, air 1–6 %, aggregates 60–75 %. Normal concrete density ≈ 2400 kg/m^3 (plain) / 2500 kg/m^3 (RCC).

2. STEP 1 – PROPORTIONING (BATCHING)

Selecting right quantities of cement, FA, CA, water & admixtures (mix design – Unit V).

- **Volume batching** : quantities by volume using gauge boxes (farma) of size 35 × 30 × 25 cm = 1 bag cement (0.035 m^3). Used for small works, nominal mixes (M5–M20). Needs **bulking correction** for sand.
- **Weigh batching** : by mass using weigh-batchers (spring, beam, electronic load cells) – **accurate**, recommended for M25 and above (IS 456 cl. 9.2.2). Tolerance : cement, water ±2 %,

aggregates ±3 %, admixtures ±3 %.

- Water measured by volume (graduated tank / flow meter) ; admixtures by weight or volume.
- Types of batching plants : manual, semi-automatic, fully automatic, computerised (RMC plants).

Nominal mixes (IS 456, Table 9)

Grade	Mix ratio (C:S:A)	Total water per 50 kg cement (L max)	Use
M5	1 : 5 : 10	60	PCC / levelling
M7.5	1 : 4 : 8	45	PCC foundations
M10	1 : 3 : 6	34	PCC, footings
M15	1 : 2 : 4	32	Plain RCC works
M20	1 : 1.5 : 3	30	General RCC

3. STEP 2 – MIXING

Aim : **uniform mix – every aggregate particle coated with cement paste.**

3.1 Hand mixing

- On a clean, water-tight platform. Sand + cement mixed dry till uniform colour, then aggregate, then water added in a **depression** gradually. Turned 3–4 times. Used only for small jobs; **add 10 % extra cement**.

3.2 Machine mixing (IS 1791 batch type mixers)

Type	Working	Use
Tilting drum	Drum axis tilts to discharge	Small–medium jobs, dry/stiff mixes
Non-tilting drum	Chute inserted for discharge	Medium jobs
Reversing drum	Blades inside, drum rotation reversed to discharge	Large works, good mixing
Pan (forced action)	Rotating blades in a fixed pan	Stiff / lean / light-weight / precast
Dual-drum, twin-shaft	Continuous / high output	RMC plants, dams

- **Order of feeding** : ¼ water + coarse aggregate → cement + sand → remaining water ; admixture along with last water.

- **Mixing time** : not less than **2 min** after all materials are in the drum (IS 456 cl. 10.3.1), at **14–20 rpm** ; over-mixing → water loss, segregation, grinding of aggregate, air loss.
- Mixer should be cleaned before each fresh batch of different mix; first batch should have 10 % extra cement.
- **Mixing water** per bag of cement must be accurate – excess water ↓ strength.

4. STEP 3 – TRANSPORTING

- Concrete should be moved **as quickly as possible** without segregation, loss of slump or contamination (initial set ~ 30 min).
- Methods : **wheel-barrows / buggies** (short, level), **manual carriers**, **dumpers / trucks**, **belt conveyors** (≤ 20°, dry mix), **chutes** (≤ 1V : 2.5H), **skip & hoist / crane & bucket** (high-rise), **transit mixers** (RMC), **concrete pumps** (horizontal up to 300 m, vertical up to 100 m +), **pneumatic placers** (tunnels).
- Pumpable concrete : slump 80–150 mm, ≥ 300 kg/m^3 cementitious, rounded aggregates, MSA ≤ ⅓ pipe dia, well-graded.
- Precautions : water-tight clean containers, avoid drop > 1.5 m, no re-tempering.

5. STEP 4 – PLACING

- Prepare : clean formwork, remove loose material, apply oil/release agent, wet the subgrade and old concrete, fix reinforcement with cover blocks, **check levels & alignment**.
- Place concrete **as near as possible to final position**, in **layers 150–300 mm** thick, each layer compacted before the previous one sets (avoid cold joints).
- **Free fall ≤ 1.5 m** (use tremie, chute, elephant trunk) ; do not use vibrators to move concrete laterally; start from the far end of formwork.
- **Underwater concreting** : tremie method, Hydrophobic pumps, bags, **prepakt (grouted aggregate)**; do not allow water flow – cement washout.
- **Pumping / Shotcrete / Preplaced aggregate** are special placing techniques.
- **Construction joints** : location at points of minimum shear (≈ ⅓ span), roughened & cleaned, cement slurry/bonding agent before next lift ; **Expansion joints** (gap 20–25 mm for lengths > 45 m) & **Contraction joints**.

6. STEP 5 – COMPACTION

Purpose : expel entrapped air (5–20 % in fresh) , make dense, avoid **honeycombing** . **1 % voids ↓ strength by 5–6 %**; 5 % voids ↓ strength 30 %.

6.1 Hand compaction

- **Rodding** (16 mm bar, pocking), **tamping** (floors, slabs), **ramming** (dry earth-moist concrete in foundations) – only for workable mixes (slump > 100 mm, upto 150 mm per layer).

6.2 Mechanical compaction – vibrators

Needle / immersion (internal) vibrator	Form / shutter (external)	Surface vibrator (screed, plate)	Table vibrator
Most common, 25–75 mm dia 3000–12000 rpm, RC members	Fixed on formwork thin, congested, precast	Slabs, roads, pavements depth ≤ 150 mm	Precast units / lab cubes

- **Immersion vibrator** : insert vertically, quickly, withdraw slowly (≈ 3 cm/s); **radius of action ≈ 6–10 × needle dia (≈ 0.3–0.6 m)** ; spacing ≈ 1.5 × radius; time 5–15 s per point until air bubbles stop & paste glistens; penetrate 100 mm into previous layer; keep ≥ 75 mm from forms (avoid bar vibrating).
- **Over-vibration** → segregation, bleeding, laitance, **honeycombing** if under-vibrated.
- Frequency : 3000–12 000 vibrations/min ; amplitude 0.5–2.0 mm. Resonance with grains for high frequency.
- **Other methods** : **centrifugation / spinning** (pipes, poles), **pressure/ vacuum dewatering** , **jolting / shock tables** , **self-compacting concrete** (no compaction).

7. STEP 6 – FINISHING

- Operations on the surface of slabs / floors to produce a **smooth, level, durable surface** .
- **Screeding** : strike-off with straight edge to the level. **Floating** (wood / magnesium float) after bleed water disappears – seals surface. **Trowelling** (steel trowel) – smooth, dense surface. **Brooming** : skid-resistant finish for pavements.
- **Power floating** (large floors), **vacuum dewatering** (removal of 15–20 % water from slab : ↑ early strength, abrasion), **dry shake hardeners** , **exposed aggregate** , **stamped / coloured** finishes.
- **Never sprinkle dry cement** on surface – crazing. Do not finish while bleed water is on

surface.

- Formed (vertical) surfaces : rubbing, bush-hammering, sand-blasting, plastering (not recommended).

8. STEP 7 – CURING

Curing = maintaining satisfactory **moisture & temperature** in concrete in first days after placement so cement hydrates fully. Without curing, strength may be only 50 – 60 % of the potential and surface becomes porous.

- IS 456 cl. 13.5 : keep moist **≥ 7 days** for OPC; **≥ 10 days** with mineral admixtures / PPC ; **≥ 14 days** in hot dry weather (or for ordinary Portland with pozzolana & ...). Start curing after the final set (within 24 h), formwork loosening not earlier than prescribed stripping times.

Methods of curing

Method	Description	Use
Ponding / immersion	Dykes of mortar, water ponded on surface	Slabs, floors, roofs
Spraying / fogging	Fine mist or sprinkling; continuous	Vertical walls, columns
Wet covering	Wet hessian, gunny bags, sand, saw dust, straw	Columns, steps
Membrane (curing compound)	Spray resin / wax / PVA / bitumen emulsion forms water-tight film; white pigmented reflects heat	Pavements, where water scarce, large areas
Polythene / plastic sheet	Impermeable sheet seals moisture	Slabs, precast
Steam curing	Low-pressure steam 40–80 °C, 12–18 h at atmos. pressure; high pressure **autoclave** 8–12 atm	Precast, factory elements, ↑ 28-d strength in 1 day
Hot water / infra-red / electrical	Accelerated curing (IS 9013) for quick strength test of cubes	Precast, test

- Effect of hot/cold weather : concreting in **hot weather (>40°C)** – retarders, cool water, early morning ; **cold weather (<5°C)** – hot water, covers, accelerators, heating; must not allow freezing before 3.5 MPa.

9. FRESH CONCRETE – WORKABILITY

Workability (IS 6461 Part 7) ★

"Property of fresh concrete which determines the **ease and homogeneity** with which it can be **mixed, placed, compacted and finished**." Broad components : **consistency / mobility** (flow), **compactability** (ease of removing air), **stability** (resistance to segregation & bleeding / cohesiveness).

- Workability is a **combined property** – no single test measures it fully.
- **Useful internal work** (energy for full compaction) + **internal friction** between particles (and with forms/reinforcement).
- Required workability depends on **member (thin, congested rebars), compaction method, placing method.**

Placing condition (IS 456 cl. 7.1)	Degree	Slump (mm)
Blinding concrete, shallow sections, **pavers**, mass concrete / **vibrated**	Very low	See CF 0.75–0.85
Mass concrete; lightly reinforced slabs, beams, walls, columns (vibrated)	Low	25 – 75
Lightly reinforced sections without vibration; heavily reinforced (vibrated)	Medium	50 – 100 / 75–100
Trench fill, in-situ piling, heavily reinforced w/o vibration	High	100 – 150
Tremie concrete, pumped	Very high	150 – 180 + (flow test)

10. FACTORS AFFECTING WORKABILITY

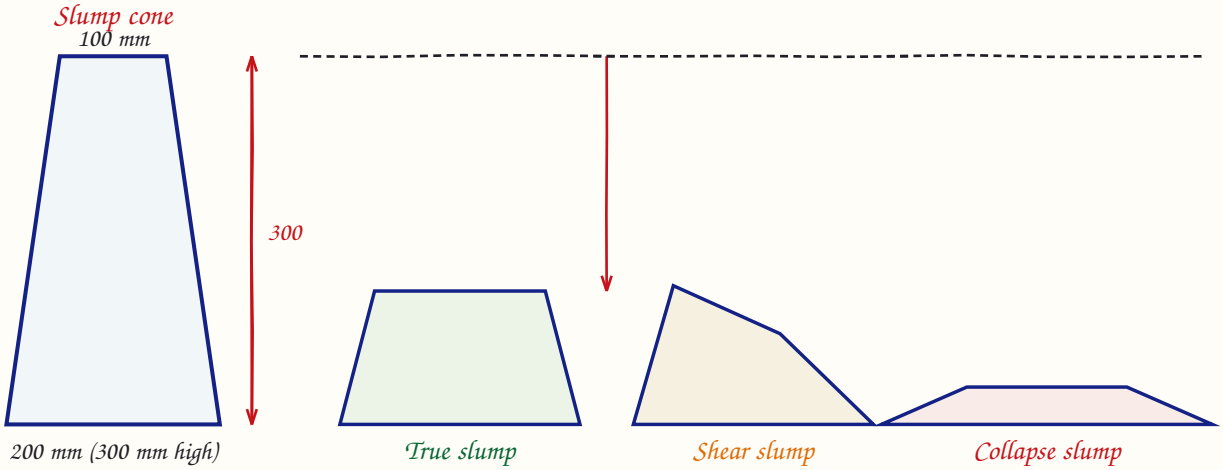
1. **Water content** (most important) : more water → higher slump; each 1 % ↑ water ≈ 6–10 mm slump; **excess water → bleeding, segregation, ↓ strength**.
2. **Cement paste** (quantity & consistency) : more paste → more lubricating → workable; very fine cement needs more water.
3. **Aggregate/cement ratio** : lean mix (more aggregate) is harsh & less workable.
4. **Aggregate size** : larger → less surface → more workable. **Shape** : rounded ≫ angular > flaky. **Texture** : smooth > rough. **Porosity** : porous aggregate absorbs water.
5. **Grading** : well-graded → fewer voids → more workable; sand content (FA %) ↑ → cohesive but less workable beyond optimum.
6. **Use of admixtures** : plasticizers, superplasticizers, air entrainers ↑ workability ; fly ash ↑ ;

silica fume ↓.

7. **Time & temperature** (slump loss), **humidity**, **wind**; **mixing time**; **cement type & fineness** (RHC less workable); **method of compaction**.

11. MEASUREMENT OF WORKABILITY

11.1 Slump test (IS 1199 : 1959) – consistency

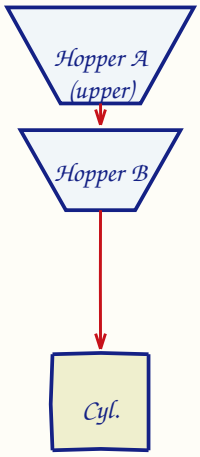


- Apparatus : frustum cone **100 mm top dia, 200 mm base dia, 300 mm high**, tamping rod 16 mm dia, 0.6 m long, bullet-nosed ; base plate, scale.
- Procedure : cone on non-absorbent plate, damp → fill in **3 layers** (each 1/3 height) → each tamped **25 times** → strike off → lift cone vertically in 5–10 s → measure subsidence (to nearest 5 mm) from top of cone.
- **True slump** : uniform subsidence – valid ; **shear slump** : one half slips – repeat test, indicates harsh/non-cohesive mix ; **collapse** : too wet / high water (too high a slump – not used for > 175 mm).
- Merits : simple, cheap, portable, used at site. Demerits : not suitable for very dry (< 10 mm) or very wet; does not measure compactability ; **does not differentiate between mixes of same slump with different aggregate.**

Slump values (typical) ★

Very low 0–25 mm : roads (vibrated) · Low 25–75 : foundations, mass concrete · Medium 50–100 : slabs, beams, columns · High 100–150 : heavily reinforced sections · Flowing > 150 mm with SP.

11.2 Compaction factor test (Glanville) – compactability

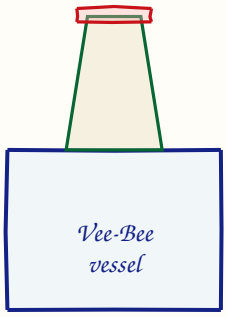


Compaction factor (Glanville)
= wt. partially compacted / wt. fully compacted
Trap doors released → concrete falls A → B → cylinder
CF 0.78 very low · 0.85 low · 0.92 medium · 0.95 high

- Suitable for **low workability** (slump 10–50 mm) – CF range **0.7 – 0.95** ; more sensitive than slump. Not suitable for CF > 0.92 (very wet) as concrete sticks in hoppers.
- Dimensions : upper hopper 25 cm top dia, 12.5 cm bottom, 22.5 cm high; lower hopper 22.5 cm dia, 22.5 cm high; cylinder 15 cm dia × 30 cm.
- CF = W_1 / W_2 ; W_1 = weight of partially compacted concrete in cylinder (falling only by gravity), W_2 = weight of fully compacted concrete (same cylinder, vibrated / rodded).

11.3 Vee-Bee consistometer test (Bahrner) – very dry mixes

Vee-Bee consistometer



Procedure :
1. Slump cone placed in cylindrical vessel on vibrating table
2. Fill in 3 layers, remove cone, set transparent disc on top
3. Start vibrator + stop-watch
4. Time (s) until disc face is fully covered with mortar
= **Vee-Bee degree** (Vebe seconds)
Suits very dry mixes : 3–30 s; slump test unsuitable.

11.4 Flow table test (IS 9103 / 1199)

- Table 76 cm square, cone 25 cm base × 17 cm top × 12 cm high placed on it ; after lifting the cone **table lifted & dropped 12.5 mm, 15 times in 15 s**. Flow % = (D – 25)/25 × 100 where D = mean spread diameter in cm. **Flow 0–150 % (medium 40–60 %)**. Good for **high workability / flowing mixes**; also detects segregation.

11.5 Other tests

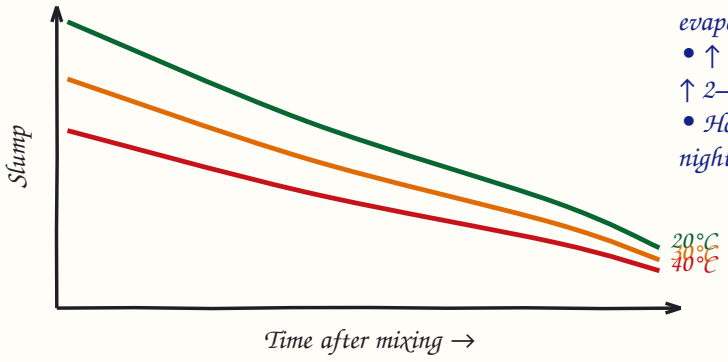
- **Kelly ball test** (ASTM C360) : 15 cm dia, 13.6 kg hemispherical ball dropped on surface; penetration depth ≈ ½ slump (2.5 cm penetration = 5 cm slump) ; fast, field test ; min depth 20 cm required.
- **Nasser's K-tester** (probe), **Remoulding test (Powers)**, **Penetration test**, **Ball penetration** .

Degree of workability	Slump (mm)	Compaction factor	Vee-Bee (s)
Very low	–	0.78	20 – 12 (≥ 20 s)
Low	25 – 75	0.85	12 – 6
Medium	50 – 100	0.92	6 – 3
High	100 – 150	0.95	3 – 0

12. SETTING TIME OF CONCRETE (IS 8142)

- Not same as cement setting time ; found by **penetration resistance test** on **mortar sieved through 4.75 mm sieve** from fresh concrete.
- Needle of different bearing area is pressed (≈ 25 mm) and **penetration resistance = force / area** vs elapsed time plotted:
  - **Initial set** : resistance = **3.5 N/mm²** (500 psi) ; **Final set** : **27.6 N/mm²** (4000 psi).
- Typical : initial 3–4 h, final 5–8 h ; retarders delay, accelerators / high temp reduce. It governs finishing, vibration re-start (stop before initial set) and formwork removal.

13. EFFECT OF TIME & TEMPERATURE



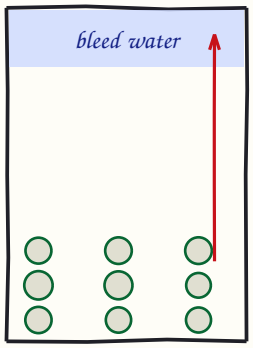
- Slump falls with time (loss of water by evaporation, absorption & hydration)
- ↑ 10 °C → slump ↓ ≈ 25 mm, water demand ↑ 2–3 %, setting time ↓ by ~ 25 %
- Hot weather : retarders, chilled water / ice, night concreting, shading, wet aggregates

- **Slump loss** : ≈ 25 – 50 mm per hour typically ; loss faster with PCC & $CaCl_2$. Hence **retempering** (adding water) not allowed (unless within limits, with SP addition).
- Remedy : add super-plasticiser at site (re-dosing), retarding plasticisers, cool materials, use

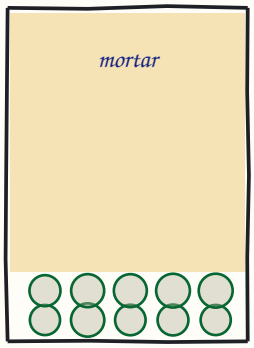
shade.

- In **cold weather** : slow hydration, set delayed ; protect against freezing & high mixing temperature (≥ 10 °C).

14. SEGREGATION AND BLEEDING



Bleeding



Segregation

Bleeding → water rises
laitance, weak top, voids
under bars/aggregate
Segregation → coarse grains
sink, mortar separates
honeycomb, weak, porous

14.1 Segregation

- **Separation of coarse aggregate from mortar** (or water from paste), making concrete **non-uniform** .
- Types : (a) coarse particles settle / separate from mix (dry mixes) ; (b) paste (cement-water) separates from aggregate (wet mixes) ; (c) water separates (bleeding).
- Causes : **too wet or too dry mix**, poorly graded / gap-graded aggregate, high MSA, **high fall (>1.5 m)**, over-vibration, rough handling, chutes with steep slope, long conveyor drop.
- Effects : honeycombing, weak spots, porosity, poor bond, erratic strength.
- Remedies : **good grading, adequate fines, lower w/c, air entrainment, limit fall height, proper vibration, SP + VMA**.

14.2 Bleeding

- **Water rising to the surface** after compaction, because solids settle (cement, 3.15 : water 1) and water is squeezed out; rate 0.1–0.5 % ; **bleeding capacity** 2–4 % of mix water.
- Causes : excess water, **low fines**, coarse cement (low specific surface), low temperature, over-vibration, deep sections, lean mixes.
- Effects : **laitance** (weak scum layer on top), porous, weak top surface (poor abrasion), poor bond with reinforcing bars & aggregates (pockets of water below them), **plastic settlement cracks**, reduces strength ≈ 10 – 15 %; sometimes beneficial – prevents plastic shrinkage by

replenishing surface water.

- Control : **finer cement** (more C_3A, alkalis), reduce w/c, ↑ fines (filler / fly ash), **air entrainment**, adequate mixing, re-vibration at proper time (**revibration**).
- Test (IS 1199 / ASTM C232) : measure volume of bleed water in 10 L container.

Difference ★

Segregation = separation of **aggregates from the mortar**; Bleeding = separation of **water from the solids** (a special form of segregation).

15. MIXING AND VIBRATION – IMPORTANT POINTS

- **Mixing** : IS 456 → mix for ≥ 2 min ; mixers must have revolution counter, water tank ; **initial batch** with extra cement ; do not use re-mixed concrete after initial setting ; time from mixing to placing **≤ 30 min** (mix) and ≤ initial set.
- **Vibration** : immersion type 25–75 mm needle dia, frequency 12 000 rpm ; avoid contact with reinforcement ; vibrate ≤ 20 s at a point ; **revibration** within permissible time ↑ strength 10–15 % & bond.
- **Needle spacing** ≈ 1.5 × radius of action ; formwork must be water-tight.
- **Safe stripping times (IS 456 cl. 11.3)** : vertical sides 16–24 h ; slab soffits (props left) 3 d ; beam soffit 7 d ; slab props 7 d (≤ 4.5 m span) ; slab soffit with props 14 d ; beams 14 d (spans > 6 m: 21 d).

16. READY MIXED CONCRETE (RMC) – IS 4926

- Concrete **produced in a central batching plant** (computerised) & delivered in **transit mixers (agitators)** to site in plastic state.

Type	Description
Central-mixed	Fully mixed at plant, hauled in agitator trucks (slow agitation) or non-agitating trucks
Transit-mixed (truck-mixed)	Dry batched at plant, mixed in truck drum enroute (70–100 revolutions at mixing speed 6–18 rpm)
Shrink-mixed	Partially mixed in plant mixer, mixing completed in truck

- Advantages : **better quality control**, speed, economy at large volume, less labour, less storage space & dust, wastage, **high strength (SP, SF) achievable**, environment.
- Limitations : transport delays/traffic ; slump loss (use retarders/SP) ; cost for small quantity;

need for pump & trained labour; plant location.

- Delivery : discharge within **2 h** of adding water ; max haul time depends on temp. Drum capacity 4–9 m^3. Tests : slump & cube strength at delivery point.

17. SHOTCRETE (GUNITE) – IS 9012

- Mortar / concrete **pneumatically projected at high velocity** (90–120 m/s) onto a surface – **self-compacting by impact**, no formwork.
- **Dry-mix process** : dry cement + sand/ aggregate (≤ 10 mm) conveyed by compressed air (0.3–0.5 MPa) through hose ; **water added at the nozzle** ; Rebound loss 15 – 50 %.
- **Wet-mix process** : ready-mixed wet concrete pumped to nozzle, air added at nozzle for projection ; lower rebound (5–15 %), more output, better quality.
- Accelerators (aluminate, silicates) for quick set on walls / overhead ; steel mesh / fibres for reinforcement ; **Rebound** = material bouncing off — avoid trapping in new layers.
- Properties : high early strength (≥ 20 MPa in 28 d, up to 50 MPa), low w/c (0.35–0.45), dense, good bond.
- Applications : **tunnel linings, slope stabilisation, canals, swimming pools, repair of concrete structures, domes, shells, rock support.**

Quick revision – Unit II ★

- Order : batching → mixing (≥ 2 min) → transport → placing (fall ≤ 1.5 m) → compaction (vibrator) → finishing → curing (7 / 10 days).
- Tests : slump (3 layers × 25), CF = W_1/W_2 (0.78–0.95), Vee-Bee seconds, flow table; setting 3.5 & 27.6 N/mm^2 (IS 8142).
- Segregation (aggregate from mortar) ≠ bleeding (water rises → laitance).
- RMC (central, transit, shrink) ; shotcrete (dry & wet mix).

CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – III

HARDENED CONCRETE

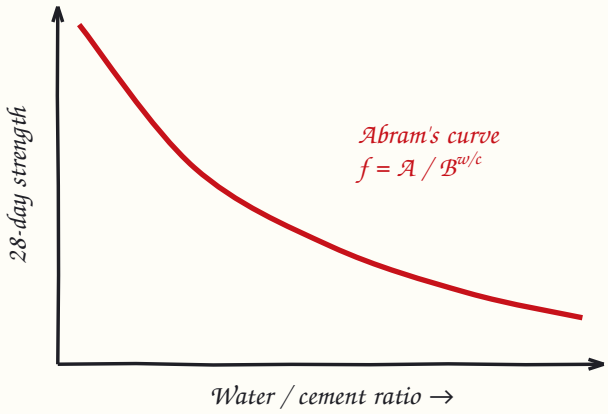
Syllabus covered in this unit

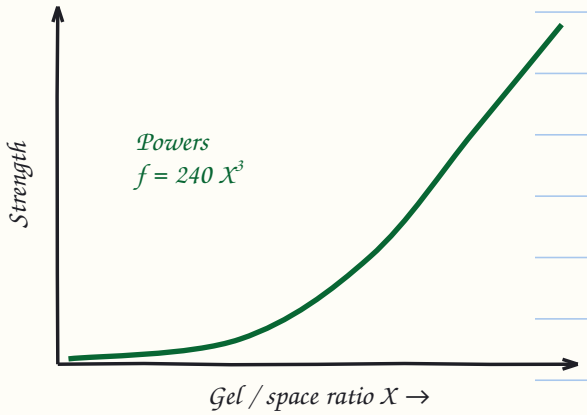
Water/cement ratio – Abram's law – gel/space ratio – nature of strength – maturity concept – strength in tension & compression – factors affecting strength – relation between compressive & tensile strength – curing – testing of hardened concrete : compression, tension, flexure, splitting – non-destructive testing – codal provisions

1. WATER / CEMENT RATIO

- **w/c ratio** = weight of water / weight of cement in the mix. It is the **single most important factor controlling strength, permeability and durability** of concrete.
- Water needed for hydration ≈ 0.23 ; for filling gel pores ≈ 0.15 → **0.38** for complete hydration. Excess water leaves **capillary pores** after evaporation → porosity ↑, strength ↓.
- If w/c < 0.38 : full hydration impossible (unhydrated cement acts as filler) but still strength high with SP; for w/c > 0.38 : capillary pores remain.
- Practical range : 0.25 (HPC) – 0.65 ; **IS 456 limits max w/c by exposure** :

Exposure	Min. grade RCC	Min. cement (kg/m^3) RCC	Max w/c (RCC)
Mild	M20	300	0.55
Moderate	M25	300	0.50
Severe	M30	320	0.45
Very severe	M35	340	0.45
Extreme	M40	360	0.40





1.1 Abram's law (D.A. Abrams, 1918)

Abram's water–cement ratio law

"For given materials and workable mix, the strength of fully compacted concrete at a given age is **inversely proportional to the w/c ratio**."

$f_c = A / B^{\chi}$ (χ = w/c by volume) Abram : $f_c = 14\,000 / 7^{\chi}$ psi

$A \approx 96$ MPa, $B \approx 4$ (SI, by weight) – A, B depend on cement, age, curing, aggregate.

- Plot : f_c vs w/c is a **hyperbola** ; **log f_c vs w/c is a straight line** ($\log f = \log A - \chi \log B$).
- **Valid** for workable, **fully compacted** mixes, same materials & curing ; fails for very low w/c (poor compaction) and for very high cement contents.
- Other relations : **Feret's** $f = K[c / (c + w + a)]^2$ (c, w, a = absolute volumes of cement, water, air) ; **Bolomey** $f = K(c/w - 0.5)$.
- **Limitation** : not consider air voids, quality of aggregate, age, etc. – Feret's is better.

2. GEL / SPACE RATIO (POWERS)

- **Gel/space ratio** X = volume of hydrated cement paste (gel) / (volume of gel + capillary pores) = **solid products ÷ available space.**

Powers' formula

$X = 0.657\,\alpha / (0.319\,\alpha + w/c)$ Strength $f_c = 240\,X^3$ (N/mm^2)

α = degree of hydration (0 – 1). Gel volume = 2.06 × vol. of cement hydrated. (Max gel/space ratio = 1.0 → 240 MPa theoretical)

- Example : w/c = 0.5, $\alpha = 0.8 \rightarrow X = 0.657 \times 0.8 / (0.319 \times 0.8 + 0.5) = 0.5256 / 0.7552 =$ **0.696** ; $f = 240 \times 0.696^3 =$ **81 MPa** (lab-paste, ≈ gel-space theory gives higher than 28-d concrete, which also depends on aggregate, ITZ, etc.).

- Strength increases as **capillary porosity ↓ (w/c ↓, α ↑)**; w/c = 0.38 → X max (1.0) when α = 1.
- Powers' theory compares **different ages & w/c on a common basis**; (Abram's law considered only w/c).

3. NATURE OF STRENGTH OF CONCRETE

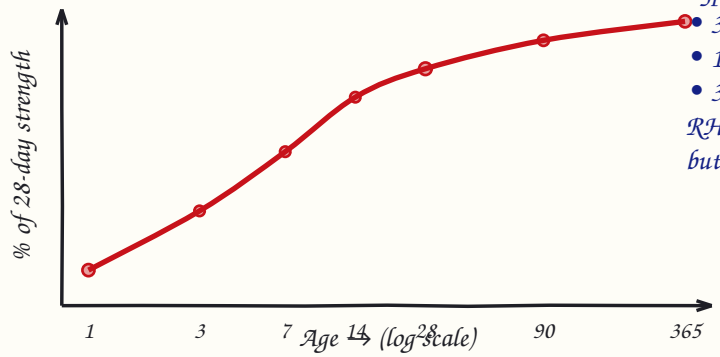
- Strength = capacity to **resist stress without failure** – compressive is the main strength ; tensile ≈ 8 – 12 % of it.
- Sources of strength : (i) **cohesion & adhesion** of hardened cement gel (**Van der Waals' forces**, chemical bonding) ; (ii) friction and **bond between paste and aggregate** (ITZ) ; (iii) mechanical interlock of aggregate.
- Concrete is a **three-phase composite** : aggregate – paste – **ITZ**; ITZ ($Ca(OH)_2$ rich, porous, micro-cracked) is the **weakest link** → micro-cracks exist even before loading (bond cracks).
- **Failure mechanism** (compression) : up to ~30 % of ultimate – stable, no new cracks ; 30–50 % – bond cracks grow ; 50–75 % – **mortar cracks** start & join ; 75–100 % – unstable crack propagation → **hourglass / cone-shear failure** (lateral tensile strain).
- Strength of the paste & aggregate > concrete strength → **ITZ governs** ; hence silica fume (densifies ITZ) ↑ strength.

4. FACTORS AFFECTING STRENGTH OF CONCRETE

1. **w/c ratio** : lower → higher strength (Abram).
2. **Cement** : type (RHC, OPC 53), fineness, compounds (C_3S, C_2S), **cement content** (strength ↑ with richness upto optimum).
3. **Aggregates** : strength, **maximum size** (small better for high strength), shape (angular ↑ flexural), **texture** (rough ↑ bond), **grading** (↑ density), cleanliness.
4. **Degree of compaction** : 5 % voids → ~30 % strength loss.
5. **Curing** : moisture + temperature ; 7-day cured gets ~ 90 % of 28-day wet-cured (OPC) ; air-cured loses about 40 %.
6. **Age** – strength increases with age if moisture available (see curve).
7. **Temperature** : high early temperature → faster strength gain but lower long-term ; below 5 °C slow ; freezing before 3.5 MPa disastrous.
8. **Admixtures & pozzolans** : plasticizers ↑ (lower w/c) ; fly ash ↓ early ↑ later ; silica fume

↑↑ ; air entrainment ↓ 5 % per 1 % air ; $CaCl_2$ ↑ early.

9. **Mixing water quality ; mixing time & method ; rate of loading** (faster → higher test strength) ; **specimen size, shape & moisture condition** (wet specimen weaker 10–15 %).
10. **Size effect** : larger specimen gives lower strength (cylinder 150×300 ≈ 0.8 × cube ; 100 mm cube ≈ 1.05 × 150 mm cube).



Typical OPC gain with age (moist cured)
• 3 d ≈ 40 % • 7 d ≈ 65 %
• 14 d ≈ 90 % • 28 d = 100 % (reference)
• 3 months ≈ 110 % • 1 year ≈ 120 %
RHC gains faster; PPC / slag slower initially but reaches higher ultimate strength

5. MATURITY CONCEPT

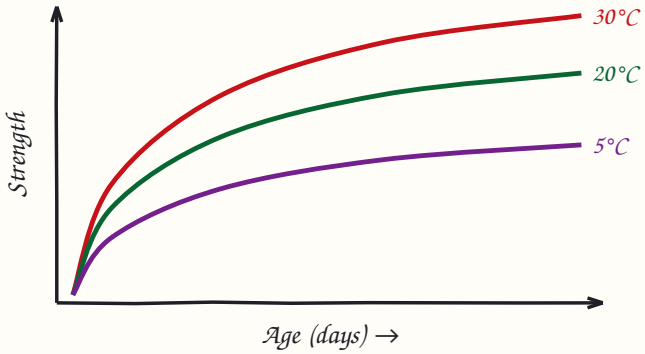
- **Maturity** = the extent of strength development as a function of **age and curing temperature** ; concrete of the same maturity has the same strength regardless of its temperature–time history (for same mix).

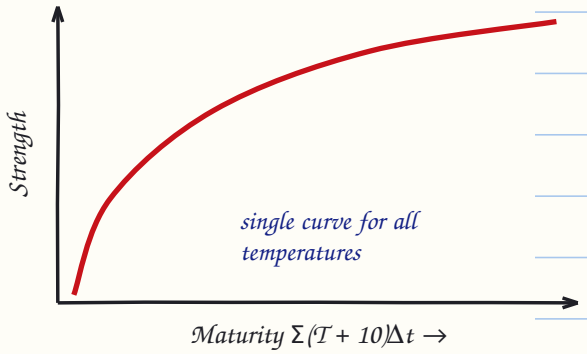
Nurse–Saul maturity function

$M = \Sigma\ (T - T_0)\,\Delta t$ (°C·h or °C·days) ; T_0 = datum temperature = −10 °C (some codes −11 °C or 0 °C)

Strength $S = a + b \cdot \log_{10}(M)$ (Plowman) ; Equivalent age $t_e = \Sigma\ (T - T_0)\,\Delta t\ /\ (T_r - T_0)$

- Example : concrete cured at 20 °C for 28 days : $M = (20 + 10) \times 28 \times 24$ = **20 160 °C·h** – concrete at 10 °C reaches same maturity in 20 160/(20×24) = 42 days.
- Uses : estimating strength **in-situ** from temperature records (thermocouples), deciding **de-shuttering / prestressing** time, cold weather concreting, steam-cured precast, mass concrete.
- Limitations : ignores initial curing temperature effects, moisture; strength–maturity curve must be calibrated for each mix; accuracy ±10–15 %.





6. STRENGTH IN COMPRESSION

- **Characteristic strength** (f_{ck}) = strength below which not more than **5 %** of test results are expected to fall = f_{mean} **– 1.65 s**.
- Grades (IS 456) : **M10, M15, M20, M25, M30, M35, M40 … M80** (M = mix, number = f_{ck} in MPa at 28 days on 150 mm cubes). Ordinary M10–M20; standard M25–M55; high strength M60–M80.
- **Design strength** $f_d = f_{ck} / \gamma_m = 0.67\ f_{ck} / 1.5 =$ **0.446** f_{ck} (limit state).
- Stress–strain curve of concrete: parabolic upto 0.002 strain then flat to $\varepsilon_{cu} =$ **0.0035** (IS 456).
- **Factors** : w/c, cement, aggregate, age, compaction, curing (see section 4).

7. STRENGTH IN TENSION

- Direct tension is difficult to apply (grip problems, eccentricity). Hence 3 indirect tests :
  - **Flexural (modulus of rupture)** $f_{cr} = 0.7\sqrt{f_{ck}}$ (IS 456 cl. 6.2.2)
  - **Split tensile** $\approx 0.56 – 0.66\sqrt{f_{ck}}$ (ACI : $0.56\sqrt{f'c}$) – slightly higher than direct
  - **Direct tension** $\approx 0.33\sqrt{f'_c}$ (ACI) – lowest.
- Why low tensile strength : micro-cracks at ITZ propagate rapidly under tension (brittle).
- Importance : **cracking load** of beams, slabs on grade, pavements, prestressed concrete (cracking moment), shear & bond.

Relation between compressive and tensile strength

Type	Formula (N/mm^2)	For M30
Flexural f_{cr} (IS 456)	$0.7\sqrt{f_{ck}}$	3.83
Split tensile f_{sp}	$\approx 0.5 – 0.6\sqrt{f_{ck}}$	2.7 – 3.3

Type	Formula (N/mm^2)	For M30
Direct tension f_t	$\approx 0.35 – 0.5\sqrt{f_{ck}}$	1.9 – 2.7
Order	$f_c \gg f_{cr} > f_{sp} > f_t$	$f_{cr} / f_{ck} \approx$ 10–15 %

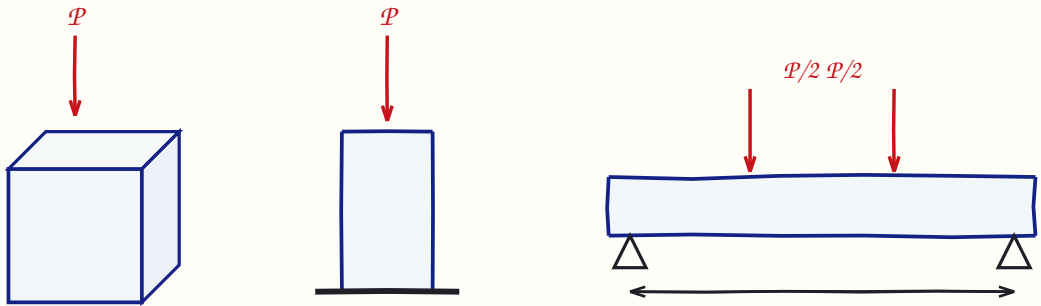
- Ratio decreases as compressive strength increases (a non-linear relation $f_t \propto f_c^{2/3}$). Tensile strength more sensitive to **curing, w/c, aggregate texture**.

8. CURING (EFFECT ON STRENGTH)

- Curing = keeping concrete moist & warm after placement to ensure hydration ; **continuous curing for 28 days gives max strength**.
- **Without curing** the strength at 28 d may be ≈ 50 % of that of properly cured concrete; 3 days wet curing → 60–70 %; 7 days → 90 %; 14 days → 95 %.
- Curing also reduces shrinkage cracking, permeability & improves **durability, abrasion resistance**.
- **Curing methods** (ponding, spraying, membrane, steam) are covered in Unit II.
- Cold weather curing > 5 °C ; hot weather – prevent evaporation (> 1 $kg/m^2/h$ → plastic shrinkage cracks).

9. TESTING OF HARDENED CONCRETE

9.1 Compressive strength test (IS 516 : 1959)

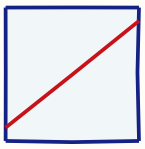


- **Specimen** : cube **150 × 150 × 150 mm** (100 mm if MSA ≤ 20 mm) or cylinder **150 dia × 300 mm** (IS 516) ; mould of cast iron / steel, oiled.
- **Casting** : fill in **3 layers** (≈ 50 mm each), **35 strokes per layer** with 25 mm square steel bar (or vibrate on table) ; level, cover with plate ; **demould after 24 ± ½ h**.
- **Curing** : in clean water at **27 ± 2 °C** until test (**3, 7, 28 days**); tested in **saturated surface-dry** state.

- **Testing** : place cube **sideways** (cast faces not in contact) in CTM, **load at 14 N/mm^2/min (= 140 kg/cm^2/min = 0.25 N/mm^2/s)** till failure ; $f_c = P / A$.
- **Minimum 3 specimens** per sample ; mean = result ; if one value differs > ± 15 % from the median it is rejected (IS 516 : individual variation ≤ ± 15 % of mean).
- Cube vs cylinder : $f_{cyl} \approx 0.8\ f_{cube}$ (for normal-strength concrete).



Satisfactory (hour-glass)



Unsatisfactory (shear / side)

Beam failure position (a = distance of crack from nearer support, b = 150 mm)
- a ≥ 133 mm : $f_{cr} = P \cdot l / (b \cdot d^2)$
- 110 ≤ a < 133 mm : $f_{cr} = 3P \cdot a / (b \cdot d^2)$
- a < 110 mm : discard result

9.2 Flexure test (IS 516)

- Beam **150 × 150 × 700 mm** (100 × 100 × 500 if MSA < 20) ; cured in water, tested **wet**.
- **Two-point (third-point) loading**, span **600 mm** (for 150 mm beam) – pure bending in middle third → shear free zone ; load rate **0.7 N/mm^2/min** (400 kg/min for 150 beam).
- **Modulus of rupture** (if crack in middle third, a ≥ 133 mm) :

Flexural strength (third-point loading)

$f_{cr} = P \cdot l / (b \cdot d^2)$ (a > 133 mm) $f_{cr} = 3\ P \cdot a / (b \cdot d^2)$ (110 < a < 133 mm)

Centre-point loading : $f_{cr} = 3\ P\ l / (2\ b\ d^2)$ P = max load (N), l = span, b, d = width & depth

- Gives **higher value than direct tension** (1.5–2×) because of non-linear stress and stress gradient ; used for **pavement slabs** design.

9.3 Splitting tensile test (Brazilian test – IS 5816)

- Cylinder **150 mm dia × 300 mm long** laid horizontally between plates (hardboard / plywood strips 3–4 mm thick, 15 mm wide) in CTM ; compression applied along a **diameter** → a nearly uniform **horizontal tensile stress** develops across the vertical diameter plane and the cylinder splits along it.

Split tensile strength

$f_{sp} = 2\ P / (\pi\ D\ L)$ Rate of loading 1.2 – 2.4 N/mm^2/min (split tensile stress rate)

P = max load, D = diameter (150 mm), L = length (300 mm) → e.g. P = 200 kN → f_{sp} = $2 \times 200000 / (\pi \times 150 \times 300) = 2.83\ N/mm^2$

- Advantages : **simple, same cylinders as for compression, more uniform than flexure**, close to direct tension (5–12 % higher).

9.4 Direct tension – briefly

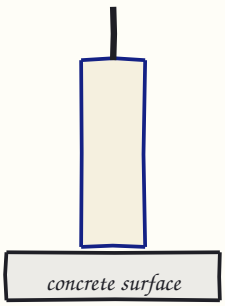
- Dog-bone / briquette specimens with embedded steel grips; hardly used because of alignment (eccentricity) problems.

10. NON-DESTRUCTIVE TESTING (NDT)

- NDT evaluates **strength, quality and defects** of in-situ concrete **without damage** ; used for old structures, cube failures, uniformity, crack depth, corrosion, repairs. **Core cutting** is semi-destructive (IS 516 Part 4).
- IS 456 cl. 17.9 : if cubes fail, NDT per **IS 13311 (Part 1 & 2)** & core test (**IS 516**); core strength ≥ 85 % of fck (avg) and no single core < 75 %.

10.1 Rebound (Schmidt) hammer & UPV

Schmidt rebound hammer



Rebound hammer (IS 13311 Pt 2) – surface hardness
- Spring-driven mass strikes plunger; rebound index R
- 9–15 readings per spot, reject outliers; R vs fck chart
- Orientation correction; affected by smoothness, moisture, carbonation, aggregate type, age
- Quick, cheap; accuracy ± 25 %

Ultrasonic Pulse Velocity (IS 13311 Pt 1)
- $V = L / t$; transducers 54 kHz, couplant grease
- >4.5 km/s excellent · 3.5–4.5 good · 3.0–3.5 doubtful · <3.0 poor

- UPV factors : moisture ↑ velocity, steel bars (↑), cracks/voids (↓), aggregate type, path length ≥ 100 mm (≥ 3 × MSA), temperature ; **Dynamic modulus** $E_d = \rho V^2 (1+\mu)(1-2\mu)/(1-\mu)$; transmission modes **direct, semi-direct, indirect (surface)**.
- **Combined method** (SONREB : rebound + UPV) – better correlation, reduces effect of moisture/age.

10.2 Other NDT methods

Method	Principle	Use
Penetration resistance (Windsor probe)	Steel probe fired by powder charge ; depth of penetration ∝ strength	Surface & near-surface strength
Pull-out test (Lok test)	Force needed to pull embedded steel insert with conical failure ≈ shear strength	Strength, de-shuttering time
Pull-off test	Tensile bond of overlays / repairs	Bond strength
Core test	100/150 mm dia core cut by diamond drill, ratio h/d = 2 (correction factors if 1–2), end-ground, tested saturated	Direct in-situ strength
Half-cell potential (ASTM C876)	$Cu/CuSO_4$ electrode vs steel : > −200 mV (90 % no corrosion) ; < −350 mV (90 % corrosion)	Reinforcement corrosion
Carbonation depth	Phenolphthalein spray : pink = uncarbonated (pH>9)	Depth of carbonation
Cover meter / GPR	Electromagnetic induction / radar	Rebar location, cover, spacing
Radiography , gamma rays	Absorption of X/γ-rays	Voids, honeycomb, steel detailing
Impact-echo / Acoustic emission / Infrared thermography	Stress waves, noise, heat patterns	Cracks, delaminations
Resistivity, chloride tests	Electrical resistivity, acid-soluble Cl content	Durability

11. CODAL PROVISIONS (IS 456)

- **Sampling frequency (cl. 15.2)** : 1–5 m^3 : 1 sample ; 6–15 : 2 ; 16–30 : 3 ; 31–50 : 4 ; > 50 : 4 + 1 per additional 50 m^3 or part thereof.
- **Test specimen** : 3 cubes per sample (7-day for early check, 28-day for acceptance).

Acceptance criteria – IS 456 cl. 16.1 (Table 11) ★

- **Mean of group of 4 non-overlapping consecutive results** ≥ f_{ck} + 0.825 s (s = established SD, min. as per Table 8) or f_{ck} + 3 N/mm^2 (M20 and above), whichever is greater.
- **Each individual result** ≥ f_{ck} − 3 N/mm^2 (M20 and above).
- If fails → NDT, core test ; **cores** : average ≥ 85 % f_{ck}, individual ≥ 75 % f_{ck} ; else structural load test (cl. 17.6) at 28 days + .

- **Assumed standard deviation (Table 8)** : M10, M15 – 3.5 ; M20, M25 – 4.0 ; M30–M50 – 5.0 N/mm^2.

- **Codes for NDT** : IS 13311 Pt 1 (UPV), Pt 2 (rebound hammer), IS 516 (Part 5 – NDT of hardened concrete), IS 456 cl. 17.9 ; ASTM C805 (rebound), C597 (UPV), C900 (pull-out), BS 1881 Part 201–207.

12. ACCELERATED STRENGTH TESTING (IS 9013)

- Purpose : predict 28-day strength within 24–28 hours for quality control & mix adjustment.
- **Warm water method** : cubes cured at 55 °C ± 2 for 20 h ; **boiling water method** : 3.5 h boiling after 23 h ; **autogenous** curing at 82 °C. Result correlated to 28-day strength by a pre-established line : $f_{28} = a + b \cdot f_{acc}$.
- Used in RMC plants for fast feedback, not for acceptance.

13. DURABILITY LINK AND PERMEABILITY

- Lower w/c → lower **permeability** (k falls from 10^{-9} to 10^{-13} m/s as w/c falls 0.7 → 0.4) → better resistance to sulphates, chlorides, carbonation, freeze–thaw, abrasion.
- Strength is usually a good index of durability **but** high strength concrete can still crack by thermal / shrinkage effects ; always check cover & curing.
- **Carbonation** : $CO_2 + Ca(OH)_2 \rightarrow CaCO_3$; pH falls from 12.5 to < 9 → steel passivity lost (depth ∝ $\sqrt{t}$).
- **Chloride limit** (IS 456 Table 7) : total acid-soluble Cl^- ≤ 0.6 kg/m³ for RCC ; sulphate (SO_3) ≤ 4 kg/m³.

14. SOLVED NUMERICALS

Problem 1 – cube strength

A 150 mm cube fails at **720 kN**. Find compressive strength. Cylinder strength?

Solution

$f_{cube} = P / A = 720 \times 10^3 / (150 \times 150) =$ **32.0 N/mm²** ; $f_{cyl} \approx 0.8 \times 32 =$ **25.6 N/mm²**

Problem 2 – Flexural strength

Beam 150×150×700, span 600 mm, third-point loading, fails at **12 kN** with crack at 200 mm from support.

Solution (a = 200 mm > 133 mm)

$f_{cr} = P\,l / (b\,d^2) = 12\,000 \times 600 / (150 \times 150^2) = 7.2 \times 10^6 / 3.375 \times 10^6 =$ **2.13 N/mm²**

Problem 3 – Using Abram's law

For a mix, A = 96, B = 4 : find f_c for w/c = 0.45 and 0.60.

Solution

$f_c = 96 / 4^{w/c}$: w/c = 0.45 → $4^{0.45} = 1.866$ → **51.4 MPa** ; w/c = 0.60 → $4^{0.60} = 2.297$ → **41.8 MPa**

Problem 4 – Maturity

Concrete cured at 15 °C for 10 days then 25 °C for 5 days. Find maturity ($T_0 = -10$ °C) and equivalent age at 20 °C.

Solution

$M = (15 + 10) \times 10 \times 24 + (25 + 10) \times 5 \times 24 = 6000 + 4200 =$ **10 200 °C·h**

Equivalent age at 20 °C = 10 200 / ((20 + 10) × 24) = **14.2 days**

Problem 5 – UPV

Path length 300 mm, transit time 66 µs. $V = 0.3 / 66\times10^{-6} =$ **4545 m/s = 4.55 km/s →** **excellent quality** concrete.

Problem 6 – Split tensile

Cylinder 150 × 300 mm fails at 180 kN : $f_{sp} = 2 \times 180\,000 / (\pi \times 150 \times 300) =$ **2.55 N/mm²**.

15. IMPORTANT QUESTIONS (EXAM)

1. State and explain Abram's law. What are its limitations?
2. Explain gel/space ratio & derive its relation with strength (Powers).
3. Discuss the factors affecting compressive strength of concrete.
4. Explain the maturity concept with an example. Where is it used?
5. Describe the procedure of the flexure test and split tensile test with formulae.
6. List NDT methods ; explain rebound hammer & UPV with their limitations.
7. Relationship between compressive strength and tensile strength of concrete.
8. IS 456 acceptance criteria for concrete cubes.

Quick revision – Unit III ★

- Abram $f = A/B^{w/c}$; Powers $f = 240\,X^3$, $X = 0.657\alpha/(0.319\alpha + w/c)$; maturity $M = \Sigma(T+10)\Delta t$.
- Cube 150 mm, 3 layers × 35, 27 ± 2 °C, load 14 N/mm²/min ; flexure $f_{cr} = Pl/bd^2$; split $f_{sp} = 2P/\pi DL$; $f_{cr} = 0.7\sqrt{f_{ck}}$.
- NDT : rebound hammer, UPV (>4.5 km/s excellent), pull-out, core, half-cell.

CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – IV

ELASTICITY, CREEP & SHRINKAGE

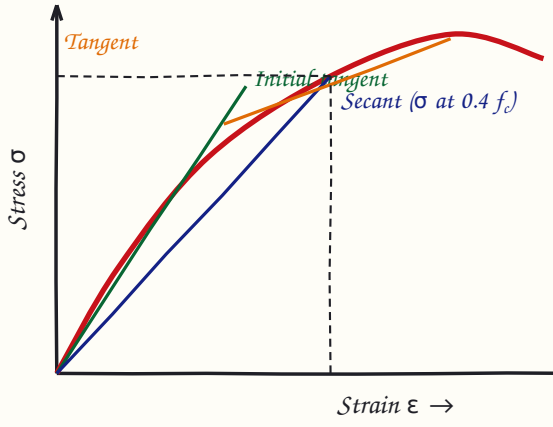
Syllabus covered in this unit

Modulus of elasticity – dynamic modulus of elasticity – Poisson's ratio – creep of concrete – factors influencing creep – relation between creep & time – nature of creep – effects of creep – shrinkage – types of shrinkage

1. ELASTICITY OF CONCRETE

- Concrete is **not truly elastic** : stress–strain curve is **curved (non-linear) from the beginning** because of its heterogenous (aggregate–paste–ITZ) nature, micro-cracking and creep. It is approximately elastic at low stress (< 30–40 % f_c).
- Elastic properties are needed for **deflection, crack width, stress in steel (modular ratio), prestress loss, column shortening, structural analysis.**
- Strains in concrete = elastic + creep + shrinkage + thermal.

1.1 Modulus of elasticity



Types of modulus of elasticity
- **Initial tangent** – slope at origin
- **Tangent** – slope at a given σ
- **Secant** – slope of line from origin to a point (≈ 0.33–0.4 f_c) – used in design
- **Chord** – slope between two points (σ = 50 μ strain to 0.4 f_c)

IS 456 : $E_c = 5000 \sqrt{f_{ck}}$ (N/mm^2)

Modulus of elasticity (Young's) E = stress / strain

Secant modulus (short-term static) is used in design (IS 456 cl. 6.2.3.1) :

$E_c = 5000 \sqrt{f_{ck}}$ N/mm^2 (ACI : $4700 \sqrt{f'_c}$; BS : $20 + 0.2\, f_{cu}$ kN/mm^2)

e.g. M20 → 22 360 ; M25 → 25 000 ; M30 → 27 386 ; M40 → 31 623 N/mm^2

- **Modular ratio** $m = E_s / E_c = 280 / (3\, \sigma_{cbc})$ (IS 456 WSM) ; for long-term including creep,

$E_{ce} = E_c / (1 + \theta)$.

- **Static modulus** = from loading test on 150 × 300 cylinder using compressometer; load–unload 2–3 cycles to remove initial non-linearity; **rate 0.25 N/mm²/s** (IS 516).
- Dependence : increases with f_c (but not proportionally) ; ↑ with age ; ↑ with aggregate stiffness (E_{agg} 20–100 GPa) ; ↓ with ↑ w/c, porosity, moisture (wet concrete slightly higher) ; ↑ with rate of loading.

Concrete grade	f_{ck} (N/mm²)	$E_c = 5000\sqrt{f_{ck}}$ (N/mm²)
M20	20	22 360
M25	25	25 000
M30	30	27 386
M35	35	29 580
M40	40	31 623
M50	50	35 355

- **Composite model** : E_c lies between Reuss (series) and Voigt (parallel) bounds of aggregate (E_a, volume V_a) & paste (E_p, V_p) :

Hirsch's / bounds

Parallel (Voigt) : $E_c = V_p E_p + V_a E_a$ Series (Reuss) : $1 / E_c = V_p/E_p + V_a/E_a$

Hirsch model : $1/E_c = \chi [1/(V_p E_p + V_a E_a)] + (1-\chi)(V_p/E_p + V_a/E_a)$

2. DYNAMIC MODULUS OF ELASTICITY

- Determined from **very small strains** (no creep, no micro-cracking) by **vibration or wave velocity** – corresponds to **initial tangent modulus** ($E_d \approx E_i$) ; Non-destructive (specimens can be re-used).
- **Methods** : (1) **Resonant frequency (longitudinal / flexural / torsional) – IS 516 Part 5 / ASTM C215** ; (2) **UPV** (IS 13311 Part 1).

Dynamic modulus

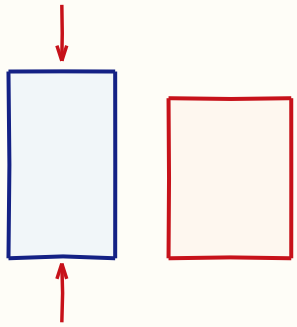
UPV : $E_d = \rho V^2 (1 + \mu)(1 - 2\mu) / (1 - \mu)$ (V = pulse velocity, ρ = density, μ = dynamic Poisson's ratio)

Resonance (longitudinal) : $E_d = 4 n^2 L^2 \rho$ (n = fundamental frequency, L = length) ; Shear $G_d = 4 L^2 n_t^2 \rho$ (torsional)

- **Relation between E_d and E_c (static)** : E_d **is higher** (≈ 20 – 40 % for lean, ~10–20 % rich/high-strength) because no creep and cracking at small strain : $E_c = 1.25 E_d - 19$ (GPa)

(Popovics) ; $E_c = 0.83\ E_d$ (typical). Higher the strength, nearer E_d to E_s.

- Useful for : **quality monitoring**, freeze–thaw damage assessment (**durability factor**), **fatigue**, earthquake analysis, effect of curing.
- Poisson's ratio can be found by combining E_d, G_d : $\mu = (E_d / 2G_d) - 1$.



← loaded shape (shorter, wider)

Poisson's ratio μ = lateral strain / axial strain
- Concrete 0.15 – 0.20 (IS 456 uses 0.2 ; creep…)
- Steel 0.28 – 0.30 ; rubber 0.5
- μ ↑ with stress near failure (> 0.75 f_c) due to micro-cracking (up to 0.5)
- Lower for high-strength & saturated concrete found 0.20 – 0.25 for UPV method

3. POISSON'S RATIO

Poisson's ratio

μ = lateral strain / longitudinal strain ; Relations $E = 2G\ (1 + \mu)$; $K = E / 3(1 - 2\mu)$

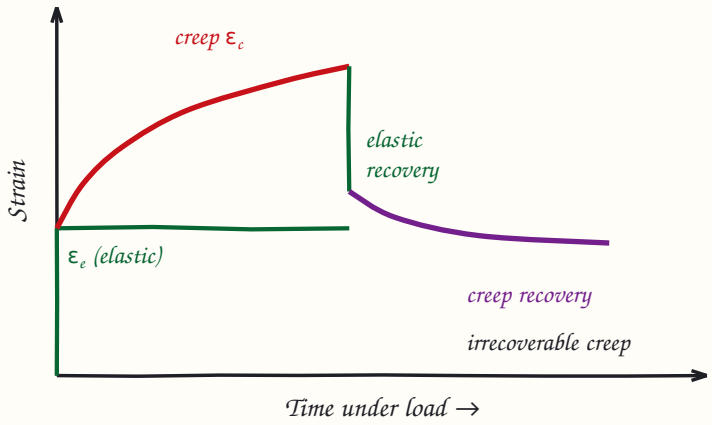
- Concrete : static 0.15 – 0.22 (average 0.2) ; dynamic 0.2 – 0.25 ; **independent of stress up to ~ 0.7 f_c**, then **increases** to ≥ 0.5 → volumetric dilation (due to internal micro-cracking) – signals imminent failure.
- Lower for high-strength concrete, dry concrete ; lightweight concrete ≈ 0.15.
- Used in analysis of slabs, tunnel linings, prestress losses, biaxial stress & pressure vessels.

4. CREEP OF CONCRETE

Definition ★

Creep = **time-dependent increase in strain under sustained (constant) stress**, after the initial elastic strain. It occurs at stress levels even well below the strength.

- Creep strain $\varepsilon_c(t)$ = total strain at t − elastic strain − shrinkage strain (a control specimen).



- Load applied at t_0 → instantaneous elastic strain ε_e
- Strain keeps increasing with time (creep)
- Rate decreases with time
- On unloading : instant elastic recovery + slow partial creep recovery + permanent (residual) deformation

Creep coefficient $\theta = \varepsilon_c / \varepsilon_e$

- **Creep coefficient** θ (φ) = $\varepsilon_c / \varepsilon_e$. **Specific creep (unit creep)** C_u = creep strain / applied stress (10^{-6} per MPa; 40 – 130 × 10^{-6}/MPa typical).
- **Creep recovery** : on unloading, instantaneous elastic recovery (~ same as elastic strain, slightly more as E rises with age) then slow creep recovery (~ 15–25 % of creep, reversible) ; the rest (**irrecoverable**) ~ 75–85 %.
- **IS 456 cl. 6.2.5.1 : creep coefficient** θ (long term) :

Age at loading	Creep coefficient θ
7 days	2.2
28 days	1.6
1 year	1.1

- Creep strain $\varepsilon_{cc} = \theta\, \varepsilon_e = \theta\, \sigma / E_c$; effective modulus $E_{ce} = E_c / (1 + \theta)$.

5. FACTORS INFLUENCING CREEP

1. **Aggregate** : stiffer & denser aggregate → lower creep ; light weight (more creep ~ 1.5×) ; crushed granite < gravel < sandstone ; **volume of aggregate ↑ → creep ↓** (aggregates restrain paste).
2. **Water–cement ratio / paste content** : creep ∝ amount of paste ; lower w/c, better compaction → less creep ; high strength concrete creeps less.
3. **Cement type & fineness** : more C_3S, rapid hardening cement → less creep (stronger at loading) ; low-heat cement more ; blended cement : fly ash/slag reduce long-term creep.
4. **Stress/strength ratio** : linear (**proportional to stress**) up to 0.4–0.5 f_c ; non-linear above (micro-cracking), ≥ 0.75 f_c → creep rupture.
5. **Age at loading** (maturity) : older concrete – less creep (θ at 7 d 2.2 vs 1 yr 1.1).
6. **Relative humidity / curing** : creep decreases as RH rises ; fully-saturated concrete (sealed)

creeps ~ 2–3 × less than dried ; **drying creep (Pickett effect)**.

7. **Temperature** : creep ↑ with temperature (60–70 °C doubles / triples).
8. **Size & shape of member** (volume / surface ratio) : thicker → less creep (**effective thickness**).
9. **Reinforcement** reduces creep by sharing load (compression steel).
10. **Admixtures** : water reducers (lower w/c ↓ creep); accelerators / air entraining ↑ ; silica fume ↓.

6. RELATION BETWEEN CREEP AND TIME

- Creep develops rapidly at first and then slows, tends to a **limiting (ultimate) value** after many years.

Time under load	% of 20-year (ultimate) creep	Remarks
2 weeks	≈ 25 %	Rapid
1 month	≈ 35 – 40 %	
3 months	≈ 50 – 55 %	
1 year	≈ 70 – 75 %	Mostly completed
5 years	≈ 85 – 90 %	
20 – 30 years	100 %	Ultimate

Empirical expressions

Ford & Hansen : $C_t = C_u \cdot t / (a + t)$ (hyperbolic ; $a \approx 10$ days, C_u = ultimate creep)

Ross : $t / C_t = a + b\,t$ ACI 209 : $C_t = [\, t^{0.6} / (10 + t^{0.6}) \,]\, C_u$ (t in days)

Log law : $C_t = a + b \log_{10}(t) - \varepsilon_c \propto \log(t)$

- Example : $C_u = 600 \times 10^{-6}$, ACI 209 at $t = 100$ days : $t^{0.6} = 15.85$; $C_{100} = 15.85/(25.85) \times 600 = 368 \times 10^{-6}$.

7. NATURE (MECHANISM) OF CREEP

- Not completely understood ; theories :
  - **Seepage (moisture-movement) theory** – squeezing out of adsorbed / gel water under stress, to capillary pores.
  - **Viscous flow of cement gel** – sliding of gel sheets / particles (amorphous mass).
  - **Delayed elasticity** – elastic aggregates restraining the creeping paste ; stress transferred slowly (recoverable portion).

- **Plastic flow** (**micro-cracking**, slip at ITZ) – at high stress; **irrecoverable** component.
- **Breakage of bonds & formation of new ones (solidification theory)**.

▶ Basic creep (sealed) + drying creep. Mostly occurs in the **cement paste** ; aggregates do not creep.

Point ★

Creep and shrinkage are similar in cause (**gel water movement**) and are **partly additive** ; they are often studied together — drying creep (Pickett).

8. EFFECTS OF CREEP

▶ **Deflection of beams & slabs** increases (long-term deflection ≈ 2–3 × short-term ; IS 456 Annex C uses factor).

▶ **Loss of prestress** in prestressed concrete (creep + shrinkage + relaxation + elastic shortening ≈ 15–25 %) → account in design.

▶ **Redistribution of stresses** : in columns, creep of concrete transfers load to the steel (steel may yield); in composite members & statically indeterminate structures moments redistribute.

▶ **Column shortening** in tall buildings (differential creep between core and columns).

▶ **Beneficial** : relieves stress concentrations (restraint stress from shrinkage & temperature), reduces cracking in mass concrete & restrained members ; relaxation of internal stresses.

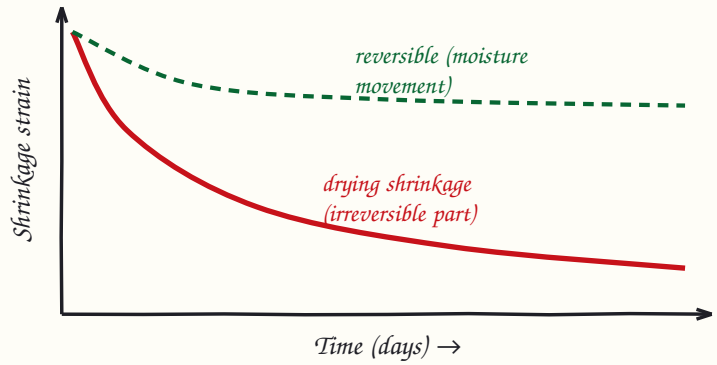
▶ **Effect on crack width, buckling** of slender columns, **arch dam** behaviour, bridges (camber loss).

▶ Remedies : lower w/c, ↑ aggregate stiffness / volume, longer curing & later loading, compression steel, higher strength, steam curing.

9. SHRINKAGE

Definition ★

Shrinkage = **volume reduction** of concrete **independent of external load**, due to loss of water / chemical & thermal changes. Cracks form if restrained.



Moisture movement
- *Drying → shrinkage ; wetting → swelling*
- *Part of drying shrinkage irreversible*
- *Typical 300–800 × 10^{-6}*

(IS 456 : 0.0003 when not available)
- *Rate ∝ exposed surface /volume*
- *Mostly in first few months*

▶ *Typical drying shrinkage* **0.0003 – 0.0008** *(3–8 × 10^{-4}) i.e. 300–800 microstrain (IS 456 cl. 6.2.4 : 0.0003 if unknown). Linear shrinkage per metre = 0.3 – 0.8 mm.*

9.1 Types of shrinkage

Type	Cause	Remarks
Plastic shrinkage	Rapid evaporation of surface water from fresh concrete (rate > 1 kg/m^2/h) before set, exceeding bleeding rate ; hot, windy, dry	Map / diagonal cracks in slabs within 1–6 h ; **prevent by fog spray, wind-breaks, curing compounds, polypropylene fibres**
Drying shrinkage	Loss of **adsorbed gel water** from hardened paste in dry air (RH < 100 %)	Main type ; partly irreversible ; proportional to water content & paste volume
Autogenous shrinkage	**Self-desiccation** – chemical shrinkage of cement hydration (hydrated products smaller than reactants by ~ 10 %), no moisture exchange	Significant for w/c < 0.42, **HPC / silica fume**, at early age
Carbonation shrinkage	$CO_2 + Ca(OH)_2 \rightarrow CaCO_3$; volume reduction (water released) at RH ≈ 50 %	Surface layer, with drying shrinkage
Thermal shrinkage	Cooling after peak hydration temperature (α ≈ 10 × 10^{-6} /°C)	Mass concrete, dams, thick raft ; control by low-heat cement, pre-cooling, pipe cooling

9.2 Factors affecting shrinkage

▶ **Water content** (↑ shrinkage ~ linearly with water content) ; **w/c ratio**, cement content ; **aggregate** (stiff, large, high volume ↓ shrinkage; sandstone, slate ↑; granite, limestone ↓) ; **fineness** & type of cement (RHC, high C_3A more) ; **admixtures** ($CaCl_2$ ↑ 10–50 %, retarders ↑, SP with water cut ↓) ; **relative humidity** (shrinkage 0 at 100 %, 4 × at 40 % RH) ; **size / shape** (V/S ratio – thicker slower) ; **curing** (longer curing delays but does not reduce) ;

reinforcement (restrains ~ 10 – 40 %) ; **carbonation** .

9.3 Control of shrinkage cracking

- Minimum water, proper aggregate grading with max size, **good curing 7–14 days** ; **contraction / control joints** (spacing 3–6 m in slabs ; < 4.5 m in walls), **temperature & shrinkage steel** (0.15 % for HYSD in slabs, IS 456 cl. 26.5.2.1), **shrinkage-compensating cement** (Type K, expansive – ettringite), **shrinkage reducing admixtures (SRA)**, fibres, fly ash, avoiding delay.

10. CREEP vs SHRINKAGE – COMPARISON

Aspect	Creep	Shrinkage
Load dependence	Under **sustained stress**	**Independent** of load
Cause	Gel water seepage, viscous flow, micro-cracking	Loss of water, thermal, chemical changes
Reversibility	Partly recoverable on unloading	Partly reversible on re-wetting
Measured by	Loaded minus unloaded (control) specimens	Unloaded specimen, strain gauge / dial gauge
Typical magnitude	2–3 × elastic strain (θ = 1.1–2.2)	300 – 800 × 10^{-6}
Time	Continues for years	Mostly in first months–1 yr
Effect	Deflection, prestress loss, stress redistribution	Cracks, warping, prestress loss

11. MOISTURE & THERMAL MOVEMENT

- **Moisture movement** : swelling (on wetting, expansion ≈ 100–200 × 10^{-6}) and shrinkage (drying) – cyclic wetting/drying causes fatigue ; **irreversible shrinkage** ≈ 30–60 % of total on first drying.
- **Coefficient of thermal expansion** of concrete : 7 – 13 × 10^{-6} /°C (typical 10 × 10^{-6}) ≈ steel (12 × 10^{-6}) → no thermal stress between steel & concrete ; depends on aggregate (quartz ↑, limestone ↓).
- **Thermal stress** if restrained : $\sigma = E \alpha \Delta T / (1 - \mu)$ (ΔT = 20 °C, E = 25 GPa, $\alpha = 10^{-5}$ → σ ≈ 5 MPa > tensile strength → cracks).

12. SOLVED NUMERICALS

Problem 1 – modulus of elasticity

Find E_c for M35 concrete, and short-term shortening of a 3 m column carrying 5 MPa.

Solution

$E_c = 5000\sqrt{35} = 5000 \times 5.916 =$ **29 580 N/mm²** ; $\varepsilon = \sigma/E = 5/29\,580 = 1.69 \times 10^{-4}$; $\Delta L = 3000 \times 1.69\times10^{-4} =$ **0.51 mm**

Problem 2 – creep deformation

Column 3 m long carries stress 6 MPa, M30 concrete, loaded at 28 days. Find long-term creep strain & total shortening ($\theta = 1.6$).

Solution

$E_c = 5000\sqrt{30} = 27\,386$ MPa ; $\varepsilon_e = 6/27\,386 = 2.19 \times 10^{-4}$; $\varepsilon_{creep} = \theta\,\varepsilon_e = 1.6 \times 2.19\times10^{-4} = 3.51 \times 10^{-4}$

Total $= 5.70 \times 10^{-4} \rightarrow \Delta L = 3000 \times 5.70\times10^{-4} =$ **1.71 mm** (creep portion 1.05 mm) ; $E_{ce} = 27\,386 / 2.6 = 10\,533$ MPa

Problem 3 – dynamic modulus

UPV = 4.4 km/s, density 2400 kg/m³, $\mu = 0.2$:

Solution

$E_d = 2400 \times (4400)^2 \times (1.2)(0.6)/(0.8) = 2400 \times 1.936 \times 10^7 \times 0.9 =$ **4.18 × 10¹⁰ Pa = 41.8 GPa** (static $E_c \approx 0.8$–$0.9\,E_d \approx 35$ GPa)

Problem 4 – shrinkage

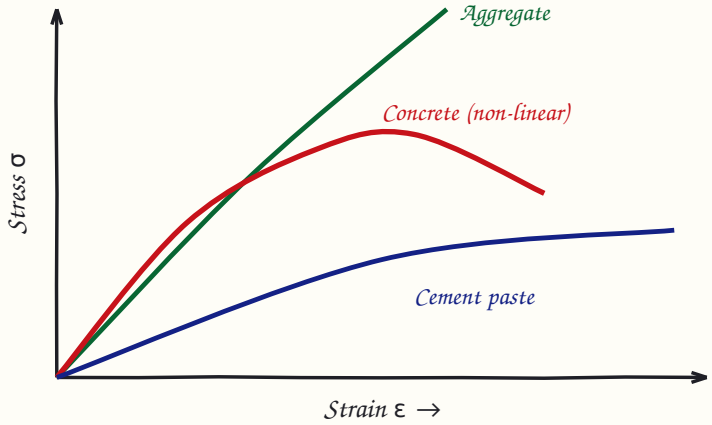
Slab 6 m long, shrinkage strain 0.0003 fully restrained, $E_c = 25\,000$ MPa : tensile stress $\sigma = E\,\varepsilon = 25\,000 \times 0.0003 =$ **7.5 MPa** (> f_{ct}, hence cracks ; creep relieves ~ 50 – 70 %) ; free shortening $= 6000 \times 0.0003 =$ **1.8 mm**.

13. IMPORTANT QUESTIONS (EXAM)

1. Define modulus of elasticity & explain types ; give IS 456 expression.
2. Explain dynamic modulus of elasticity and how it is determined. Relate it with static modulus.

3. Define creep. Explain factors influencing creep & the creep–time relationship.
4. Explain nature (mechanisms) and effects of creep on structures.
5. What is shrinkage ? Explain different types of shrinkage & factors affecting.
6. Compare creep and shrinkage. How cracking due to shrinkage is controlled ?
7. Define Poisson's ratio & state values for concrete.

14. STRESS–STRAIN BEHAVIOUR OF CONCRETE



- Aggregate and paste are each almost linear-elastic ; concrete is **curved** due to ITZ micro-cracking
- IS 456 design curve : parabola up to $\varepsilon = 0.002$ then constant to $\varepsilon_{cu} = 0.0035$
- Peak stress at $\varepsilon \approx 0.002$; HSC is steeper and more brittle
- Descending branch only in stiff machines

- Region 1 ($\sigma < 0.3\ f_c$) : nearly linear, bond cracks (pre-existing) stable ; Region 2 (0.3–0.5) : slight curvature, bond cracks grow ; Region 3 (0.5–0.75) : mortar cracks, strain increases faster ; Region 4 (> 0.75 f_c) : unstable cracking → failure at **$\varepsilon \approx 0.002$–$0.0035$** .
- **Ultimate strain** : 0.0035 (IS 456), 0.003 (ACI); HSC lower (0.0025–0.003), confinement increases ductility (stirrups, spiral).
- **Design stress block** : 0.45 f_{ck} (WSM bending) ; limit state : 0.446 f_{ck} over 0.42 x_u.

15. TESTS FOR CREEP & SHRINKAGE

- **Creep test (ASTM C512 / IS 1343)** : 150 × 300 mm cylinders in a **creep rig** (spring-loaded frame) under constant stress ≤ 40 % f_c ; companion unloaded specimens measure shrinkage; temperature 23 ± 2 °C, RH 50 ± 4 % ; strain measured with Demec (mechanical) gauge / LVDT / vibrating-wire gauges at set time intervals (1 d, 3 d, 7 d, 14 d, 28 d, 3 m, 6 m, 1 y).
- **Shrinkage test (IS 1199 / ASTM C157)** : prism 75 × 75 × 285 mm with gauge studs ; length comparator reading after 28-day moist curing then drying at 50 % RH – strain = $\Delta L / L$. Reading at 4, 7, 14, 28, 56 days etc.
- **Moisture movement** : prisms alternating wet–dry.

16. RHEOLOGICAL MODELS (BRIEF)

- **Maxwell model** (spring + dashpot in series) – stress relaxation. **Kelvin–Voigt** (parallel) – delayed elasticity / creep recovery. **Burgers model** (Maxwell + Kelvin in series) fits creep well : instantaneous elastic + delayed elastic + viscous flow.
- $\varepsilon(t) = \sigma/E_1 + (\sigma/E_2)(1 - e^{-t/\tau}) + \sigma\, t / \eta$ (Burgers).
- **Relaxation** : decrease in stress at constant strain over time – important in prestressing (concrete relaxation ≈ 20–30 % of creep stress loss).

17. DESIGN PROVISIONS (CREEP, SHRINKAGE)

- **Long-term deflection (IS 456 cl. 23.2 & Annex C)** : $\delta_{total} = \delta_{shrinkage} + \delta_{creep+instantaneous}$; deflection ≤ span/250 (total) and span/350 or 20 mm (after finishes) ; use effective modulus E_{ce}.
- **Prestress losses (IS 1343)** : creep loss = $\varepsilon_{cc}\, E_s$; ε_{cc} = ultimate creep strain per MPa (= 0.000 044 / MPa typical) × stress at steel level ; shrinkage loss = $\varepsilon_{cs}\, E_s$ (ε_{cs} = 0.0003 pre-tension, 0.0002 / $\log_{10}(t+2)$ post-tension) ; relaxation of steel 3–9 %.
- **Reinforcement for shrinkage & temperature** : minimum 0.15 % (Fe 415, 500) – 0.12 % of gross area in slabs (IS 456 cl. 26.5.2.1) ; spacing ≤ 5 d or 450 mm.
- **Joints** : expansion joints for structures > 45 m ; control joints spacing 3–4.5 m in floor slabs on grade.

18. MORE NUMERICALS

Problem 5 – prestress loss due to creep

Stress at steel level = 7 N/mm^2, specific creep = 4.4×10^{-5} per N/mm^2, $E_s = 2 \times 10^5$ N/mm^2 → creep loss = $7 \times 4.4\times10^{-5} \times 2\times10^5$ = **61.6 N/mm^2** (≈ 4.4 % of 1400 N/mm^2 initial stress).

Problem 6 – equivalent modulus ratio

Concrete M25, $E_s = 2\times10^5$. Short-term modular ratio m = 200 000 / 25 000 = **8** ; with θ = 1.6 → long-term m = 8 × 2.6 = **20.8** (shows steel stress rise in columns with time).

Problem 7 – thermal strain

A 30 m roof slab, $\alpha = 10 \times 10^{-6}/°C$, temperature rise 25 °C : free expansion = $30\,000 \times 10^{-5} \times 25 =$ **7.5 mm** – provide expansion joint.

19. ONE-PAGE SUMMARY

Property	Symbol / value	Range for normal concrete
Static modulus	$E_c = 5000\sqrt{f_{ck}}$	20 – 40 GPa
Dynamic modulus	$E_d = \rho V^2(\dots)$	25 – 50 GPa (≈ 1.1–1.4 E_c)
Poisson's ratio	μ	0.15 – 0.22
Creep coefficient	$\theta = \varepsilon_c/\varepsilon_e$	1.1 – 2.2 (IS 456)
Drying shrinkage	ε_{sh}	$300 – 800 \times 10^{-6}$
Thermal expansion	α	$7 – 13 \times 10^{-6}$ /°C
Ultimate strain	ε_{cu}	0.0035 (IS 456)

Quick revision – Unit IV ★

- $E_c = 5000\sqrt{f_{ck}}$; $E_d > E_s$; $\mu = 0.15–0.2$; $E_d = \rho V^2(1+\mu)(1-2\mu)/(1-\mu)$.
- Creep θ : 2.2 (7 d), 1.6 (28 d), 1.1 (1 yr) ; creep ∝ stress up to 0.4 f_c.
- Shrinkage : plastic, drying, autogenous, carbonation (+ thermal) ; IS 456 : 0.0003.

CONCRETE TECHNOLOGY

Subject Code : 23A01402T | Handwritten Study Notes

UNIT – V

MIX DESIGN & SPECIAL CONCRETES

Syllabus covered in this unit

Ready mixed concrete – fibre reinforced concrete (types of fibres, factors) – high performance concrete – self-consolidating concrete – self-healing concrete – factors in choice of mix proportions – quality control – statistical methods – acceptance criteria – ACI method & IS code method of mix design

1. READY MIXED CONCRETE (RMC)

- **RMC** = concrete batched at a **central plant** under controlled conditions and **delivered in plastic state** to the site by transit mixers (IS 4926 : 2003).
- Types : **central-mixed, transit-mixed, shrink-mixed** (see Unit II).
- Plant : aggregate bins & silos (cement, fly ash), weigh batchers (accuracy ± 2 %), water & admixture dosing, control room (computerised), truck mixers (4–9 m^3, 2–3 rpm agitation, 6–18 rpm mixing), pumps/boom pumps.
- **Advantages** : consistent quality (SD ≈ 3.5 N/mm^2), faster construction, less labour & storage at site, cleaner, minimal wastage, use of mineral & chemical admixtures, high-grade concrete (M30–M80).
- **Disadvantages** : traffic delay, slump loss (retarders, SP), not economical for small quantities, plant investment, need for pump.
- **Quality control** : fresh tests (slump, temperature, air) at delivery; cubes 28 d; delivery docket details (grade, slump, time of batching, quantities).
- Specification : **designed mix (producer responsible for strength)** vs **prescribed mix** (purchaser specifies proportions).

2. FIBRE REINFORCED CONCRETE (FRC)

Definition ★

FRC = concrete made of cement, aggregates (≤ 10–20 mm) and **discrete, randomly oriented short fibres** (0.1–2.0 % by volume) to improve **tensile strength, toughness, crack control and impact resistance.**

2.1 Types of fibres

Steel fibre shapes (increasing anchorage →)



Straight Crimped Hooked-end Deformed Twisted

Aspect ratio l/d = 50 – 100 ; length 20–60 mm ; dia 0.3–1 mm ; V_f 0.25 – 2 %

Fibre	Properties	Remarks / use
Steel (SFRC)	Dia 0.3–1 mm, length 20–60 mm, E = 200 GPa, f_t = 0.8–2 GPa ; V_f 0.25–2 % ; hooked-end, crimped, deformed	Pavements, industrial floors, tunnel shotcrete, precast, airfield; rust at crack surface only
Glass (GFRC, AR-glass)	Alkali-resistant (Zr-coating), E = 70 GPa, f_t = 1.7–3.5 GPa ; 2–5 % by weight ; l = 12–25 mm	Facade panels, cladding, architectural, thin sheets ; ordinary E-glass attacked by alkalis
Polypropylene (PP)	Low modulus (3.5–8 GPa), f_t 0.4–0.7 GPa, hydrophobic, chemically inert ; V_f 0.1–0.5 % ; l = 12–19 mm	**Plastic shrinkage crack control**, fire spalling resistance, shotcrete
Carbon	E = 230–400 GPa, f_t = 2–3 GPa ; costly	High strength, cladding, blast
Natural / organic	Jute, coir, sisal, bamboo, hemp, cellulose ; low cost, biodegradable ; l ≈ 25–50 mm	Low-cost roofing sheets, rural housing ; durability issue (alkali degrades)
Asbestos (obsolete)	E = 160 GPa ; health hazard	No longer used
Nylon, polyester, basalt, aramid	Elastic, impact resistant ; basalt = high temp. resistance	Specific uses

2.2 Factors affecting properties of FRC

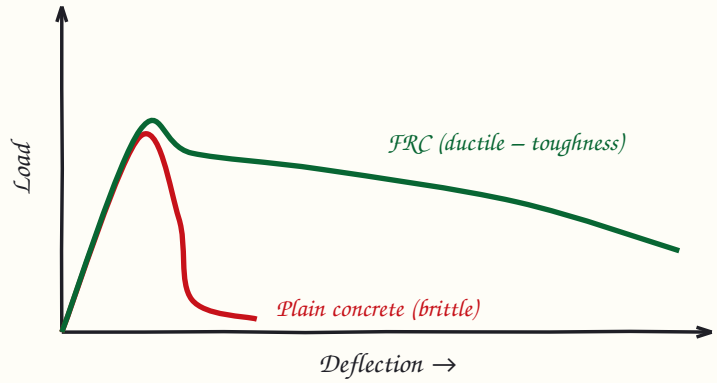
Key parameters ★

1. **Fibre type & material** (modulus, strength, bond) 2. **Aspect ratio (l/d)** 3. **Volume fraction V_f** 4. **Orientation & distribution** 5. **Matrix strength (w/c)** 6. **Aggregate size & content** 7. **Mixing, placing & compaction method** 8. **Bond (shape, surface)**

▶ **Aspect ratio** (l/d) : 50–100 ; > 100 → balling (poor mixing) ; < 30 → pull-out. Greater

aspect ratio improves strength & toughness but lowers workability.

- **Volume fraction** : up to 2 % for steel (> 2 % balling) – strength ↑ almost linearly ; effective at V_f > **critical volume**.
- **Orientation** : random (3D) efficiency factor ≈ 0.41 (2D 0.64, aligned 1.0) ; vibration orients fibres along flow ; thin section → 2D.
- **Max aggregate size ≤ 10–12 mm** (larger sizes obstruct fibre ; mortar-rich mix, cement ≥ 350–450 kg/m^3, fly ash, sand 50–70 % of aggregate).
- **Bond** : pull-out of smooth fibres dominates ; deformed / hooked ones fail by fibre fracture.
- **Mixing** : add fibres last, through vibrating screen / belt dosing, using SP ; avoid balling.



Fibres bridge cracks :
- ↑ flexural & tensile strength
- ↑↑ toughness & impact
- ↑ fatigue, shear, abrasion
- ↓ shrinkage cracks
- Compressive strength ≈ same
- Workability ↓

2.3 Properties and mechanics of FRC

- **Compressive strength** : unchanged / ↑ 0–25 % ; **flexural strength** ↑ 40–150 % (steel 1–2 %) ; **split tensile** ↑ 30–50 % ; **impact** ↑ 3–10× ; **toughness index** ↑ 10–40× ; **fatigue endurance limit** ↑ (0.65 → 0.9 f_r) ; **shrinkage** ↓ ; **durability** (freeze thaw) similar.

Fibre factor & Composite law

Fibre factor $\mathcal{F} = \beta\ (l/d)\ V_f$ (β = bond factor : 0.5 round, 0.75 crimped, 1.0 indented)

Composite $\sigma_{cu} = \sigma_m V_m + \eta\ \sigma_f V_f$

Critical fibre length $l_c = \sigma_f\ d\ /\ (2\ \tau_b)$ (τ_b = bond stress) – effective only when $l > l_c$; pull-out when $l < l_c$

Flexural strength (Swamy) : $\sigma_{cu} = 0.97\ \sigma_{cm}\ V_m + 3.41\ (\tau\ l\ /\ d)\ V_f$

- **Applications** : industrial floors / **pavements** & airport runways, **tunnel linings (shotcrete)**, canal lining, bridge decks, precast pipes, dams (repair), refractory, blast-resistant & seismic structures, thin shell, **overlays**.
- **Limitations** : workability, cost, balling, fibre corrosion at cracks (steel), nonuniform

distribution, fire (PP melts).

3. HIGH PERFORMANCE CONCRETE (HPC)

Definition (ACI) ★

HPC = concrete meeting **special combinations of performance & uniformity** (workability, strength, durability, volume stability, early strength) **that cannot be achieved routinely with conventional constituents & practices.** HSC (high-strength) ⊂ HPC. IS 456 : HSC = M65 & above.

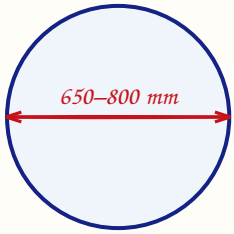
- **Characteristics** : strength 60 – 120 MPa ; very low w/b (**0.25 – 0.35**) ; high workability (slump > 200 mm with SP) ; high early strength ; high E (35–50 GPa) ; very low permeability (< 10^{-13} m/s) ; high durability (chloride, sulphate, carbonation) ; low creep & shrinkage ; **very brittle**.
- **Materials & mix** : Cement 400–550 kg/m^3 (OPC 53) ; **silica fume 5–10 %** ; fly ash / GGBS 15–40 % ; **superplasticizer (PCE) 1–2 %** ; crushed hard aggregate MSA 10–12.5 mm (strong, low absorption) ; w/b 0.25–0.35 ; VMA / SRA optional ; **cementitious total ≤ 600 kg/m^3**.
- **Mix-design approach** : (i) low w/b by SP (ii) pozzolan to fill voids & densify ITZ (iii) good aggregate (iv) optimum paste volume ; trial mixes.
- **Special care** : thorough mixing (long, 3–5 min), tight QC, **curing** is essential (early & wet 7–14 d ; internal curing by saturated lightweight aggregate / SAP for autogenous shrinkage), thermal control (high heat – cooling of mass).
- **Applications** : high-rise columns (Burj Khalifa 80 MPa), long span bridges, offshore platforms, nuclear containment, pavements, marine structures, precast / prestressed.
- **Advantages** : smaller sections ↑ usable area ; longer service life (> 100 yrs) ; lower life-cycle cost ; lower creep, shrinkage.

4. SELF-COMPACTING CONCRETE (SCC)

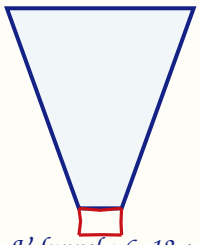
- SCC (Okamura, Japan, 1988) flows under its **own weight**, fills formwork & encapsulates reinforcement **without vibration**, while remaining homogeneous (EFNARC 2002 / IS 456 Annex /IS 10262 guidelines).
- **Key properties** : (1) **Filling ability** (flowability) (2) **Passing ability** (through congested

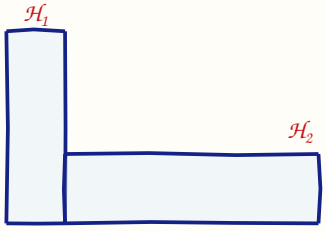
rebars) (3) **Segregation resistance** (stability).

- **Mix differences vs normal concrete** :
  - Higher **powder content** (cement + fly ash/GGBS/limestone filler/silica fume) 450–600 kg/m^3 ; **w/powder ratio 0.80–1.10 by volume** (w/p ≈ 0.28–0.40 by mass).
  - Lower **coarse aggregate** (28–35 % of concrete volume ; ≈ 750–900 kg/m^3) with MSA 12–20 mm ; sand 48–55 % of total aggregate.
  - **Superplasticizer (PCE)** (flow) + **VMA** (viscosity modifying agent – cohesion).
  - Water 150–200 kg/m^3.



Slump flow


V-funnel : 6–12 s



L-box : $H_2/H_1 \geq 0.8$


U-box : fill ≤ 30 mm

Test	Measures	Acceptance (EFNARC)
Slump flow (Abram's cone)	Filling ability, flow diameter ; T_{50} (time to 500 mm)	650 – 800 mm ; T_{50} = 2 – 5 s
V-funnel	Filling ability / viscosity	Flow time 6 – 12 s ; (V-funnel at 5 min ≤ +3 s : segregation)
L-box	Passing ability (blocking)	$H_2/H_1 \geq 0.8$
U-box	Passing ability	Height difference ≤ 30 mm
J-ring	Passing ability (with rebars)	Difference from slump flow ≤ 10 mm (≤ 50 mm acceptable)
Sieve segregation (5 mm sieve)	Segregation resistance	≤ 15 % (5–15 %), > 30 % unacceptable
Column segregation	Stability	≤ 15 %

- **Advantages** : no vibration (noise, labour), easy placing in congested sections, better surface finish & bond, faster construction, durable ; precast & RMC friendly.
- **Disadvantages** : costlier (powders, SP, VMA), need strict quality control, formwork pressure higher (full hydrostatic), sensitive to water variation.
- **Applications** : congested RC members, precast, tunnels, underwater, retrofitting, architectural, repair.

5. SELF-HEALING CONCRETE

- Concrete that **repairs its own cracks** automatically, restoring watertightness / durability (cracks < 0.1–0.3 mm for natural, up to 0.8 mm for engineered).

5.1 Autogenous healing (natural)

- Continued hydration of **unhydrated cement** (exposed in crack face) ; **carbonation** of $Ca(OH)_2$ → $CaCO_3$ crystal (dominant) ; swelling of C-S-H ; debris blocking. Needs water (wet–dry cycles) & CO_2, w/c < 0.4 (more unhydrated cement), fly ash/GGBS (late hydration) & crystalline admixtures.

5.2 Autonomous (engineered) healing

Technique	Mechanism	Remarks
Bacterial (bio-concrete)	Spores of *Bacillus pasteurii / pseudofirmus / subtilis (Sporosarcina)* + nutrient (calcium lactate) in clay pellets / capsules ; on crack & water ingress the bacteria wake & precipitate $CaCO_3$ (calcite)	TU Delft (Jonkers) ; seals 0.5–0.8 mm cracks in weeks ; viable > 200 yr (spores)
Microcapsules / polymer	Brittle capsules with epoxy / polyurethane / sodium silicate / MMA break at the crack & release healing agent	Quick, but one-time ; sizes 10–200 µm
Vascular network	Hollow glass tubes / fibres or 3D network supply healing agent from reservoir	Repeated healing ; complex, costly
Super-absorbent polymers (SAP)	Swell on water entry and block the crack; also internal curing	Sealing + extra hydration
Shape-memory polymers / ECC	SMP fibres contract when heated, closing cracks ; ECC (engineered cementitious composite) keeps crack widths < 60 µm so autogenous healing works	Research / niche
Crystalline admixture	Reactive chemicals precipitate crystals in the presence of water	Commercial (e.g. waterproofing)

- **Advantages** : longer service life, reduces maintenance cost, ↓ permeability, protects steel.
  Limitations : cost, limited crack width, viability of agents during mixing & high pH, strength reduction (capsules), standardisation lacking.

6. FACTORS IN THE CHOICE OF MIX PROPORTIONS

Mix design = determining the proportions of cement, water, FA, CA (and admixtures) to produce concrete of **required strength, workability, durability at minimum cost.**

1. **Compressive strength (target mean)** $f'_{ck} = f_{ck} + 1.65\ s$ (or $f_{ck} + X$ per IS 10262 : 2019).
2. **Workability** (placing, compaction, section thickness).
3. **Durability** – exposure condition → max w/c & min cement content (IS 456 Table 5).
4. **Type & grade of cement** (OPC 43/53, PPC), **aggregates** (MSA, shape, grading, zone, specific gravity).
5. **Maximum size of aggregate** (≤ ¼ min. member size) ; **admixtures** (SP, retarder) ; **mineral additions** (fly ash, GGBS).
6. **Quality control** – degree of supervision (standard deviation s) ; **economy** (cement content minimum).
7. Special : pumping, hot weather, mass concrete heat, shrinkage.

- **Methods** : arbitrary (1 : 2 : 4), fineness modulus, maximum density (Fuller), **ACI**, **IS 10262**, DOE (British), Road Note No. 4, Talbot, **trial mix**.

Exposure (IS 456 Table 5)	Max w/c (RCC)	Min cement (kg/m^3) & grade
Mild	0.55	300 ; M20
Moderate	0.50	300 ; M25
Severe	0.45	320 ; M30
Very severe	0.45	340 ; M35
Extreme	0.40	360 ; M40

7. QUALITY CONTROL OF CONCRETE

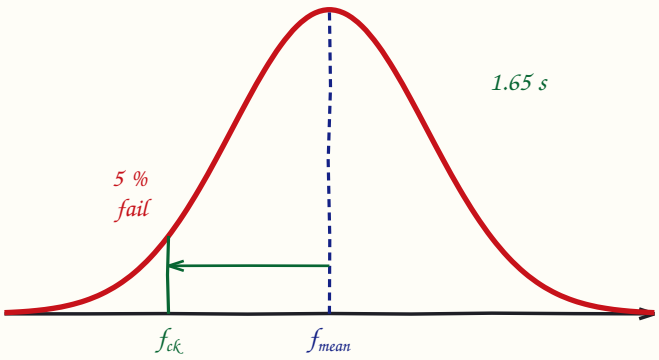
- **Quality control (QC)** = systematic procedures to ensure concrete **consistently meets specification** — covers **materials, mix proportions, batching, mixing, transport, placing, compaction, curing, testing, records.**
- **Essential points** : testing of cement (IS 4031), aggregates (IS 2386), water, steel ; calibration of batching plants ; periodic checks of **moisture content, bulking** ; slump & temperature at site ; **cube testing** (sampling frequency IS 456 cl. 15) ; maintaining **records** & control charts ; trained manpower ; supervision.
- **Variation causes** : materials (cement, aggregate grading), batching errors, mixing/handling,

testing errors (sampling, casting, curing, machine), site conditions.

Degree of control	Std. deviation s (N/mm^2)	Characteristics (ACI 214 – overall)
Excellent	< 2.8	Laboratory quality / RMC
Very good	2.8 – 3.5	Good plants
Good	3.5 – 4.2	Site with supervision
Fair	4.2 – 4.9	Average
Poor	> 4.9	Little control

- **Cost–quality** : high QC → lower s → lower target mean strength → **saving of cement** (cost).

8. STATISTICAL METHODS & ACCEPTANCE CRITERIA



Normal distribution of strength
- $f_{ck} = f_{mean} - 1.65\ s$ (5 % defectives)
- s = standard deviation
- $Cv = (s / f_{mean}) \times 100$ %
- f'_{ck} (target) = $f_{ck} + 1.65\ s$

- **Mean** $\bar{x} = \Sigma x / n$; **Standard deviation** $s = \sqrt{[\Sigma (x - \bar{x})^2 / (n - 1)]}$; **Coefficient of variation** $Cv = s / \bar{x} \times 100$ % (good < 10–15 %) ; **Range** $R \approx 2.5–3\ s$ for small samples.
- **Normal (Gaussian) distribution** of strength test results (symmetrical bell curve).

  Characteristic strength (95 % probability) $f_{ck} = f_m - 1.65\ s$; target mean strength $f'_{ck} = f_{ck} + 1.65\ s$.

Grade	s (IS 456 Table 8 / IS 10262 Table 2)	X (IS 10262 : 2019)
M10, M15	3.5	5.0
M20, M25	4.0	5.5
M30 – M50	5.0	6.5
M55 and above	6.0	8.0

- Target strength is the **higher** of ($f_{ck} + 1.65\ s$) and ($f_{ck} + X$) (IS 10262 : 2019). If s is from ≥ 30 test results, use actual s (IS 456 cl. 9.2.4).

Example

M25, $s = 4.0 \rightarrow f'_{ck} = 25 + 1.65 \times 4 = 31.6$; $25 + X = 25 + 5.5 = 30.5 \rightarrow$ target mean = **31.6 N/mm²**

Test data : 28.0, 31.5, 29.5, 33.0, 30.0 → mean 30.4 ; $s = \sqrt{[(2.4^2 + 1.1^2 + 0.9^2 + 2.6^2 + 0.4^2)/4]} = \sqrt{(14.7/4)} =$ **1.92** ; Cv = 6.3 %

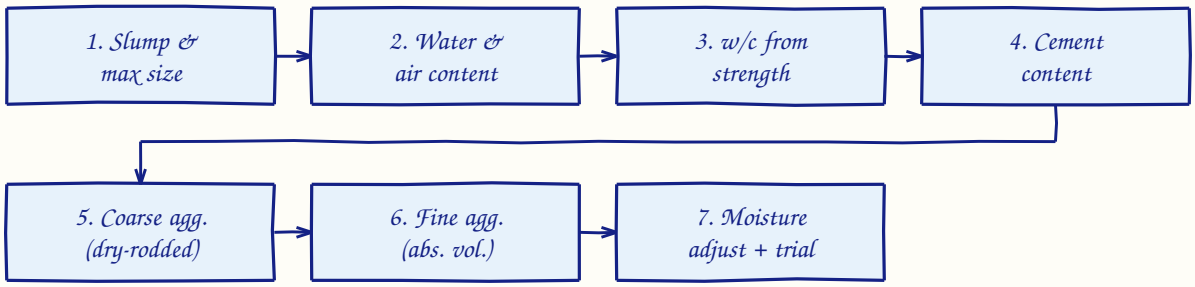
Acceptance criteria (IS 456 : 2000 cl. 16, Table 11) ★

- **Mean of any group of 4 consecutive test results** ≥ $f_{ck} + 0.825 \times s$ (established SD, rounded to nearest 0.5) or $f_{ck} + 3$ N/mm² (M20 and above), whichever is greater (for M15 and below : $f_{ck} + 0.825\ s$ or $f_{ck} + 3$, with no further rule).
- **Any individual test result** ≥ $f_{ck} - 3$ N/mm² (for M15 and below : ≥ $f_{ck} - 3$, i.e. 85 % f_{ck} for M10–M15 in older practice).
- Concrete is deemed to **comply** if both met ; else non-conforming → core / NDT / load test.

▶ **Control charts** ($\bar{x}$ & R charts) – upper/lower control limits ± 3 s help detect trends early. **Cumulative-sum (CUSUM)** also used. **Moving average of 7**.

9. ACI METHOD OF MIX DESIGN (ACI 211.1)

▶ Based on **empirical tables** & **absolute volume** method ; mix ratios per 1 m³ ; suits normal-weight concrete.



ACI 211.1 absolute-volume method – step flow

▶ **Step 1 : Slump** from Table (e.g. columns 25–75 mm, beams 25–100 mm, pavements 25–75 mm) ; **Step 2 : Max aggregate size**.

▶ **Step 3 : Mixing water** & air from tables (non-air-entrained) :

Slump (mm)	9.5 mm	19 mm	37.5 mm
25 – 50	207 kg/m³	190	166
75 – 100	228	205	181
150 – 175	243	216	190

Slump (mm)	9.5 mm	19 mm	37.5 mm
Entrapped air %	3.0	2.0	1.0

- **Step 4 : w/c** from strength (non-AE, 28-day mean) ; 45 MPa → 0.38 ; 40 → 0.42 ; 35 → 0.47 ; 30 → 0.54 ; 25 → 0.61 ; 20 → 0.69 ; 15 → 0.79. Also durability limits.
- **Step 5 : Cement content** = water / (w/c) (check ≥ minimum for durability).
- **Step 6 : Coarse aggregate** : bulk volume of dry-rodded CA per unit volume of concrete b/b_o from table (function of MSA & **FM of sand**), × dry-rodded unit weight.

Max size (mm)	FM = 2.4	2.6	2.8	3.0
9.5	0.50	0.48	0.46	0.44
19	0.66	0.64	0.62	0.60
25	0.71	0.69	0.67	0.65
37.5	0.75	0.73	0.71	0.69

- **Step 7 : Fine aggregate** = 1 m^3 – (water + cement + CA + air) volumes (absolute volume), × specific gravity. (Alternative : weight method, from estimated density of fresh concrete 2355 kg/m^3 for 19 mm).
- **Step 8 : Adjust for moisture & absorption** ; **Step 9 : trial batch**, check slump, strength, adjust.

ACI worked example

Data : target 30 MPa, slump 75–100 mm, MSA 19 mm, FM_{sand} = 2.6, G_c = 3.15, G_{FA} =

Water = **205 kg** ; air 2 % ; w/c (30 MPa) = **0.54** → cement = 205/0.54 = **380 kg**

CA = 0.64 × 1600 = **1024 kg**

Abs. volumes : water 0.205 + cement 380/3150 = 0.1206 + CA 1024/2700 = 0.3793 + air 0.020 = 0.7249 m^3 → sand volume = 0.2751 m^3 → **FA = 0.2751 × 2650 = 729 kg**

Mix per m^3 : Cement 380 : FA 729 : CA 1024 : Water 205 → **1 : 1.92 : 2.69, w/c 0.54**

10. IS CODE METHOD (IS 10262 : 2019)

- Applies IS 456 requirements ; for ordinary & standard concrete (M10–M70) with 28-day f_{ck} ; uses **w/c curves, water table & CA volume table**.

Steps ★

1. **Stipulations** : grade, MSA, cement type, exposure, workability, SG of materials, zone of sand, admixture.
2. **Target mean strength** : $f'_{ck} = f_{ck} + 1.65\ s$ or $f_{ck} + X$ (higher).
3. **w/c ratio** : from curve (cement strength vs 28-d strength) ; adopt lesser of this and IS 456 Table 5 limit.
4. **Water content** (Table 4 : 10 mm → 208, 20 mm → 186, 40 mm → 165 kg/m^3 for 25–50 mm slump, angular, Zone II) ; **+3 % per 25 mm slump above 50 mm** ; reduce 10 kg for rounded ; **SP reduces ≥ 20 %**.
5. **Cement content** = water / (w/c) ; check min (Table 5) & max 450 kg/m^3 (OPC) / up to 540 with SCM.
6. **CA volume fraction** (Table 5) : per unit volume of total aggregate for MSA & sand zone (at w/c 0.5) ; adjust ± 0.01 for each ∓ 0.05 change in w/c.
7. **Mix calculations** by absolute volume ; total aggregate volume = 1 − (cement + water + admixture volumes) ; CA & FA masses from fractions.
8. **Trial mixes** (3 trials) & adjust ; correct for moisture in aggregates.

Vol. of CA per unit vol. of total aggregate (w/c 0.5)	Zone IV	Zone III	Zone II	Zone I
10 mm	0.50	0.48	0.46	0.44
20 mm	0.66	0.64	0.62	0.60
40 mm	0.75	0.73	0.71	0.69

IS 10262 worked example (M25, moderate exposure)

Data : M25, OPC 43 (G = 3.15), 20 mm angular CA (G = 2.74), Zone II sand (G = 2.65),

Target = 25 + 1.65×4 = **31.6 MPa** ; w/c adopted = **0.46** (≤ 0.50 limit)

Water = 186 × 1.06 (50→100 mm : +6 %) = **197 kg** ; Cement = 197/0.46 = **428 kg** (≥ 300 ✓ ; ≤ 450 ✓)

CA fraction = 0.62 + (0.50 − 0.46)/0.05 × 0.01 = 0.628 ; FA = 0.372

Total aggregate vol = 1 − 428/3150 − 0.197 = 0.667 m^3

CA = 0.667 × 0.628 × 2.74 × 1000 = **1148 kg** ; FA = 0.667 × 0.372 × 2.65 × 1000 = **658 kg**

Mix : Cement 428 : FA 658 : CA 1148 : Water 197 = **1 : 1.54 : 2.68, w/c 0.46**

ACI vs IS 10262 – comparison

Feature	ACI 211.1	IS 10262 : 2019
Origin	USA, empirical tables	India, adapted from DOE / ACI, with IS 456 limits
Strength basis	28-day cylinder f'_c ($f'_{cr} = f'_c + 1.34\,s$)	28-day cube f_{ck} ; target = $f_{ck} + 1.65\,s$
Water	From slump & MSA table, air entrainment	Table 4 (+3 % / 25 mm slump) ; SP reduces
CA content	Dry-rodded volume (b/b_o) by FM of sand	CA fraction of total aggregate by zone
FA	Absolute volume / weight by difference	Absolute volume by difference
Durability	Separate exposure categories	IS 456 Table 5 limits

Quick revision – Unit V ★

- RMC : central, transit, shrink-mixed (IS 4926) ; FRC : steel/glass/PP, l/d 50–100, $V_f \le 2$ % ; HPC : w/b 0.25–0.35, SF + SP ; SCC : powder 450–600, slump flow 650–800 mm, V-funnel 6–12 s, L-box ≥ 0.8 ; self-healing : bacteria $CaCO_3$, capsules, vascular.
- $f_{ck} = f_m - 1.65\,s$; target = $f_{ck} + 1.65\,s$; acceptance : mean of 4 ≥ f_{ck}+0.825s (or +3), individual ≥ f_{ck}–3.
- ACI : water → w/c → cement → CA (dry-rodded) → FA (abs. vol.) ; IS 10262 : target → w/c → water (186 for 20 mm) → cement → CA fraction → masses.